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Biosynthetic production of gamma- and delta-lactones using cytochrome P450 enzymes having subterminal hydroxylase activity

Abstract

Provided herein are methods for making lactones comprising incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising straight chain fatty acids to produce subterminal hydroxy fatty acids, and incubating a yeast cell culture with the subterminal hydroxy fatty acids to produce lactones.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a Continuation of International Patent Application No. PCT/US2020/024071, filed Mar. 21, 2020, entitled “BIOSYNTHETIC PRODUCTION OF GAMMA-AND DELTA-LACTONES USING CYTOCHROME P450 ENZYMES HAVING SUBTERMINAL HYDROXYLASE ACTIVITY”, which claims the priority of the U.S. Provisional Application Ser. No. 62/821,894, filed on Mar. 21, 2019, the entire contents each of which are incorporated herein by reference.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

(1) The contents of the electronic sequence listing (C149770067US01-SEQ-ZJG.xml; Size: 38,878 bytes; and Date of Creation: Sep. 21, 2021) is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The field of the invention relates to methods and processes for producing gamma- and delta-lactones via modified microbial cultures. More specifically, the present disclosure relates to the production of gamma- and delta-lactones having 5-8 carbon atoms from corresponding fatty acid substrates via enzymatic conversion.

BACKGROUND OF THE INVENTION

(3) There exists a universal desire for foods, fragrances and cosmetics that have pleasant tastes and odors. In many cases, these traits of pleasant smells and tastes are provided through various lactone

compounds, including various gamma- and delta-lactones, that possess desirable aroma and flavor characteristics. Current demand for lactone compounds is mainly addressed through either chemical synthesis or extraction processes from plants. Plant extraction-based production has significant disadvantages, such as weather effects on the strength and abundance of the compounds of interest, risk of plant diseases and/or poor harvest, stability of the compound, environmental impact of increased production and trade restrictions. Industrial production can cause environmental damage, may use dangerous precursors and itself may be subject to increased cost due to cost of key substrates.

(4) Accordingly, there is a need in the art for novel methods to produce gamma- and delta-lactones economically and reliably without the limitations posed by plant extraction and chemical synthesis.

SUMMARY OF THE INVENTION

(5) According to the current invention, lactones may be reliably produced at high yield by gene modification and fermentation technology using microorganisms such as bacteria and/or yeasts. These microorganisms are able to synthesize lactones de novo or by biotransformation of fatty acids to provide commercially significant yields. Hence, new production methods are provided herein to reduce costs of gamma- and delta-lactone production and lessen the environmental impact of large-scale cultivation and processing of natural sources from which these lactone compounds can be extracted.

(6) More specifically, the present disclosure encompasses methods and compositions for making lactones by microbial fermentation, wherein the microbial fermentation comprises a cellular system expressing a heterologous cytochrome P450 (CYP450) protein.

(7) The present disclosure is based, in part, on the finding that certain CYP450 proteins and functional variants thereof have hydroxylase activity at specific subterminal carbon atom positions. In particular, certain CYP450 proteins have been identified to cause hydroxylation of straight chain fatty acids at the first (ω -1), second (ω -2), and/or third (ω -3) carbon atoms adjacent to the terminal (ω) carbon atom. By overexpressing these CYP450 proteins in a modified microbial system and feeding straight chain fatty acids as substrates to the said modified microbial system, ω -1, ω -2, and ω -3 hydroxy fatty acids are produced which then can be converted efficiently into various gamma- and delta-lactones at high titers. The present disclosure, therefore, provides economical and reliable methods for producing gamma- and delta-lactones from straight chain fatty acids without the disadvantages associated with chemical synthesis or plant extraction as described above.

(8) Accordingly, one aspect of the present disclosure provides a method for making lactones comprising (a) incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, and incubating the resulting subterminal hydroxy fatty acid with a cell culture followed by acid treatment to produce one or more gamma- and delta-lactones.

(9) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila*. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila*. In some embodiments, the CYP450 protein from *Myceliophthora thermophila* comprises the amino acid sequence of SEQ ID NO: 3.

(10) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina*. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina*. In some embodiments, the CYP450 protein from *Umbelopsis isabellina* comprises the amino acid sequence of SEQ ID NO: 6.

(11) Any methods described herein may further comprise expressing an isocitrate dehydrogenase (Icd) protein (e.g., a heterologous Icd) in the cellular system. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 90% (e.g., at least 90%, at least 91%, at least 92%, at least 93%, at least 94%) identical to the amino acid sequence of Icd from *E. coli*. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 95% (e.g., at least 95%, at least 96%, at least 97%, at least 98%, at least 99%) identical to the amino acid sequence of Icd from *E. coli*. In some embodiments, the Icd protein from *E. coli* comprises the amino acid sequence of SEQ ID NO: 4.

(12) In some embodiments, the straight chain fatty acid comprises 5 carbon atoms to 40 carbon atoms. In preferred embodiments, the straight chain fatty acid comprises 7 carbon atoms to 20 carbon atoms. In some embodiments, the fatty acid is selected from lauric acid, tridecylic acid, or a combination thereof. In some embodiments, the subterminal hydroxy fatty acid comprises hydroxylated lauric acid, hydroxylated tridecylic acid, or a combination thereof. In some embodiments, the subterminal hydroxy fatty acid comprises a hydroxyl group at a position selected from the group consisting of ω -1, ω -2, and ω -3.

(13) In some embodiments, the cellular system can comprise growing cells. In some embodiments, the cellular system can comprise lyophilized cells. In some embodiments the cellular system can comprise bacterial cells, yeast cells, plant cells that do not naturally produce the lactone of interest, algal cells, fungal cells, or combinations thereof, one or more of which have been transformed to overexpress the protein(s) described herein. In some embodiments, the cellular system can comprise bacteria and/or yeast cells selected from the group consisting of *Escherichia*; *Salmonella*; *Bacillus*; *Acinetobacter*; *Streptomyces*; *Corynebacterium*; *Methylosinus*; *Methylomonas*; *Rhodococcus*; *Pseudomonas*; *Rhodobacter*; *Synechocystis*; *Saccharomyces*; *Zygosaccharomyces*; *Kluyveromyces*; *Candida*; *Hansenula*; *Debaryomyces*; *Mucor*; *Pichia*; *Torulopsis*; *Aspergillus*; *Arthrobotrys*; *Brevibacteria*; *Microbacterium*; *Arthrobacter*; *Citrobacter*; *Klebsiella*; *Mucor*; *Pantoea*; *Corynebacterium*; and *Clostridium*.

(14) In some embodiments, the lactones can comprise five carbons to eight carbons. In some embodiments, the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone. In some embodiments, the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.

(15) In some embodiments, the straight chain fatty acid is lauric acid, the subterminal hydroxy fatty acid is hydroxy lauric acid, and the lactones comprises one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone. In some embodiments, the straight chain fatty acid is tridecylic acid, the subterminal hydroxy fatty acid is hydroxy tridecylic acid, and the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone. In some embodiments, the subterminal hydroxy fatty acid is isolated from the cellular system prior to treating it with cell culture.

(16) Any methods described herein may further comprise isolating the lactones from the microbial cell culture or the acidic environment to provide a crude product. In some embodiments, the crude product obtained from such isolating step can include a lactone content that is at least 70% pure. In some embodiments, the method further comprises: i) purifying a crude product comprising the lactones; and, ii) removing solvents under vacuum to provide a concentrated product. In some embodiments, said crude product is purified by column chromatography. In some embodiments, said crude product is purified by acid-base extraction. In some embodiments, said crude product is purified by vacuum distillation. In some embodiments, the method further comprises purifying said lactones using a semi-preparative HPLC.

(17) The disclosure further provides a fermentation product composition comprising a mixture of lactones, wherein the mixture of lactones comprises one or more gamma- and/or delta-lactones having 5 to 8 carbon atoms. The lactone content in the fermentation product composition can be at least 70% (e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at

least 95%, at least 97%, at least 98%, at least 99% or 100%) of the crude mixture and then can be further purified out of the microbial cell culture.

(18) One skilled in the art will recognize that the lactone composition(s) produced by the methods described herein can be further purified and mixed with other lactones, flavors, or scents to obtain a desired composition for use in a variety of consumer products or foods. For example, a lactone composition described herein can be included in food products (such as beverages, soft drinks, ice cream, dairy products, confectioneries, cereals, chewing gum, baked goods, etc.), dietary supplements, medical nutrition, as well as pharmaceutical products to give desired flavor or aromatic characteristics. In some embodiments, the disclosure provides a consumable product comprising a flavoring or scented amount of a lactone produced by any one of the methods above or described elsewhere herein for use in a specific flavoring context, formula or recipe. In some embodiments, the consumable product can be a food, beverage, perfume, or cosmetic product. In some embodiments, the composition can be selected from beverages, confectioneries, bakery products, cookies, and chewing gums. In certain embodiments, the composition can be a food, beverage, perfume, or cosmetic product having a flavor or aroma profile designed to taste or smell like: peach, apricot, pear, maple, coconut, tropical, butterscotch, grenadine and/or date.

(19) While the disclosure is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawing and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

(20) Other features and advantages of this invention will become apparent in the following detailed description of preferred embodiments of this invention, taken with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings form part of the present specification and are included to further demonstrate certain aspects of the present disclosure, which can be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

(2) FIG. 1 shows a schematic diagram illustrating an embodiment of the biosynthetic pathway for producing gamma- and delta-lactones according to the present disclosure, in which a C18 fatty acid (stearic acid) is bioconverted into delta-hexalactone via a 2-step process involving the subterminal hydroxy fatty acid (17-hydroxy stearic acid) as an intermediate.

(3) FIG. 2A shows GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C12:0 from the corresponding straight chain fatty acid C12:0 in a cellular system in which *M. thermophila* CYP505A30 and *E. coli* *icd* were overexpressed. Samples were taken 4 hours after bioconversion.

(4) FIG. 2B shows GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C13:0 from the corresponding straight chain fatty acid C13:0 in a cellular system in which *M. thermophila* CYP505A30 and *E. coli* *icd* were overexpressed. Samples were taken 4 hours after bioconversion.

(5) FIG. 3 illustrates the production of δ -hexalactone (DC6), γ -hexalactone (GC6), and 8-octalactone (DC8) from lauric acid (C12:0).

(6) FIGS. 4A, 4B, 4C, 4D, and 4E show GC/MS spectra confirming production of gamma- and delta-lactones from subterminal hydroxy fatty acids HO-12:0. FIG. 4A is at the top. FIG. 4B is the

second from top. FIG. 4C is in the middle. FIG. 4D is the fourth from top and FIG. 4E is at the bottom. Peaks are identified based on the retention times and mass spectra of the peaks compared with standards from Sigma. FIG. 4B and peak 1: γ -hexalactone (GC6); FIG. 4C and peak 2: δ -hexalactone (DC6); FIG. 4D and peak 3: phenethyl alcohol; and FIG. 4E and peak 4: δ -octalactone (DC8).

(7) FIG. 5 illustrates the production of γ -valerolactone (GC5), δ -heptalactone (DC7), and γ -heptalactone (GC7) from tridecanoic acid (C13:0).

(8) FIGS. 6A, 6B, 6C, and 6D show GC/MS spectra confirming production of gamma- and delta-lactones from subterminal hydroxy fatty acids HO-13:0. Peaks are identified based on the retention times and mass spectra of the peaks compared with standards from Sigma. FIG. 6A is at the top. FIG. 6B is the second from top. FIG. 6C is the third from the top and FIG. 6D is at the bottom. FIG. 6B and peak 1: phenethyl alcohol; FIG. 6C and peak 2: γ -heptalactone (GC7); FIG. 6D and peak 3: δ -heptalactone (DC7).

(9) FIGS. 7A and 7B show GC/MS spectra confirming production of subterminal hydroxy fatty acid HO-C12:0 (peak 1 in FIG. 7A) and HO-C13:0 (peak 1 in FIG. 7B) from C12:0 and C13:0, respectively, in a cellular system in which the gene MI2 was overexpressed. Samples were taken 9 h after the bioconversion and silylated with N-Triethylsilyl-N-methyl trifluoroacetamide.

(10) FIGS. 8A and 8B show GC/MS spectra confirming lactone production from different hydroxy fatty acids. FIG. 8A: *Candida boidinii* culture grown in MI2-C12-CSL medium. FIG. 8B: *Candida boidinii* culture grown in MI2-C13-CSL medium. Abbreviations in FIGS. 8A-8B have the following meaning: GC6: γ -hexalactone; DC7: δ -heptalactone (DC7); C12: lauric acid; C13: tridecanoic acid. Samples were taken 2 days after inoculation.

DESCRIPTION OF THE PREFERRED EMBODIMENT

(11) Lactones are industrially important flavor compounds that are widely found in foods, fruits, and beverages and are used in many fruity aromatic foods and cosmetics (see, e.g., Romero-Guido et al., 2011. Biochemistry of lactone formation in yeast and fungi and its utilization for the production of flavor and fragrance compounds. Appl. Microbiol. Biotechnol. 89:535-547). The present disclosure provides, in some embodiments, methods of making lactones from fatty acids. In some embodiments, the method comprises a first step, in which a cellular system expressing a heterologous cytochrome P450 (CYP450) protein converts fatty acids to hydroxy fatty acids. In some embodiments, the method comprises a second step, in which a microbial cell culture, e.g., a yeast cell culture, shorten the length of subterminal hydroxy fatty acids through a biochemical process called β -oxidation to appropriate length which upon cyclization yield appropriate lactones. Thus, the length of the subterminal hydroxy acids at the end of the second step of the process according to the present invention is shorter than the length of the subterminal hydroxy fatty acids obtained at the end of the first step of the process according to the present invention. In some embodiments, the second step further comprises an acidification step which induces the cyclization of the shortened subterminal hydroxy fatty acids to appropriate lactones. Methods provided herein further comprise isolating the lactones from the cell culture or the acidic environment.

(12) Methods of Making Lactones

(13) Methods described herein, in some embodiments, provide for the production of lactones from fatty acids. In some embodiments, methods of making lactones comprise (a) a step of incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, (b) a step of incubating the subterminal hydroxyl fatty acid from step (a) with a microbial cell culture to shorten the length of the subterminal hydroxy acid to appropriate length and (c) a step of subjecting the shortened subterminal hydroxy acid from step (b) to cyclization in an acidic environment to obtain appropriate lactone.

(14) In some embodiments, methods of making lactones comprise incubating cell pellets harvested from the cellular system expressing the heterologous CYP450 protein with a medium comprising

fatty acids. In some embodiments, methods of making lactones comprise incubating resuspended cell pellets harvested from the cellular system with the medium comprising fatty acids.

(15) Methods of making lactones described herein comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids at any ratio. In some embodiments, methods of making lactones comprise incubating a cellular system comprising between 1 gram of cells per liter to 200 grams of cells per liter (1 g/L to 200 g/L) and a medium comprising between 1 gram of fatty acids per liter to 20 grams of fatty acids per liter (1 g/L to 20 g/L). In some embodiments, methods of making lactones comprise incubating a cellular system comprising 100 grams of cells per liter (100 g/L) and a medium comprising 3 grams of fatty acids per liter (3 g/L).

(16) In some embodiments, methods of making lactones comprise incubating a cellular system with a medium comprising fatty acids for a suitable period sufficient for formation of hydroxy fatty acids. In some embodiments, methods of making lactones comprise contacting the hydroxy fatty acids with a cell culture for a suitable period sufficient for formation of lactones in the microbial cell culture. In some embodiments, methods of making lactones comprise isolating the hydroxy fatty acids from the cellular system prior to contact with the microbial cell culture. In some embodiments, methods of making lactones comprise incubating the hydroxy fatty acids in a medium with the microbial cell culture.

(17) Methods of making lactones described herein encompass incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids for a specific period of time. In some embodiments, methods of making lactones comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids for a period of time between 0.5 hour to 96 hours.

(18) Methods of making lactones described herein encompass incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids at a specified temperature. In some embodiments, methods of making lactones comprise incubating a cellular system expressing a heterologous CYP450 protein with a medium comprising fatty acids at a temperature of 16° C. to 40° C.

(19) Cytochrome P450

(20) Cytochrome P450 (CYP450) proteins, generally, are enzymes that catalyze oxidation of substrates. CYP450 proteins are members of the hemeprotein superfamily, and contain heme as a cofactor. CYP450 enzymes have been identified in many organisms including, but not limited to, animals, plants, fungi, bacteria, archaea, and viruses. Any CYP450 protein that catalyzes subterminal hydroxylation of fatty acids may be used in methods described herein, or any functional variant or fragment thereof. Examples of CYP450 proteins that have been reported to have subterminal hydroxylase activity include, but are not limited to, CYP102A1 (or P450.sub.BM3) from *Bacillus megaterium*, CYP450 protein from *Myceliophthora thermophila* (Uniprot #G2QDZ3), CYP450 from *Fusarium oxysporum*, CYP450 protein from *Umbelopsis isabellina*, CYP147G1 from *Mycobacterium marinum*, CYP109D1 from *Sorangium cellulosum*, CYP709C1 from *Triticum aestivum*, CYP109B1 from *Bacillus subtilis*, CYP505D6 from *Phanerochaete chrysosporium*, or any functional variant or fragment thereof. In some embodiments, more than one CYP450 protein is used in a method described herein. In some embodiments, 2, 3, 4, or 5 different CYP450 proteins are used in a method described herein.

(21) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila* encoded by the nucleic acid sequence of SEQ ID NO: 1. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Myceliophthora thermophila* encoded by the nucleic acid sequence of SEQ ID NO: 1. In some embodiments, the CYP450 protein comprises an amino acid sequence encoded

by nucleic acids in SEQ ID NO: 1.

(22) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Myceliophthora thermophila* provided in SEQ ID NO: 3. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Myceliophthora thermophila* provided in SEQ ID NO: 3. In some embodiments, the CYP450 protein comprises the amino acid sequence of SEQ ID NO: 3.

(23) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina* encoded by the nucleic acid sequence of SEQ ID NO: 5. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Umbelopsis isabellina* encoded by the nucleic acid sequence of SEQ ID NO: 5. In some embodiments, the CYP450 protein comprises an amino acid sequence encoded by nucleic acids in SEQ ID NO: 5.

(24) In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of CYP450 from *Umbelopsis isabellina* provided in SEQ ID NO: 6. In some embodiments, the CYP450 protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of CYP450 protein from *Umbelopsis isabellina* provided in SEQ ID NO: 6. In some embodiments, the CYP450 protein comprises the amino acid sequence of SEQ ID NO: 6.

(25) CYP450 proteins, in some embodiments, catalyze conversion of fatty acids to hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -1 to ω -3 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -1 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -2 hydroxy fatty acids. In some embodiments, the CYP450 protein catalyzes formation of ω -3 hydroxy fatty acids.

(26) Isocitrate Dehydrogenase (Icd)

(27) Isocitrate dehydrogenase (Icd) proteins, generally, are enzymes that catalyze oxidative decarboxylation of isocitrate. Icd proteins have been identified in many organisms including, but not limited to, animals, plants, fungi, bacteria, archaea, and viruses. Icd proteins, in some embodiments, convert NAD.sup.+ to NADH or NADP.sup.+ to NADPH. In some embodiments, NADH and/or NADPH generated by an Icd protein may be utilized by another enzyme (e.g., CYP450), thus promoting the activity of the other enzyme. Any Icd protein that catalyzes conversion of NAD.sup.+ to NADH or NADP.sup.+ to NADPH may be used in methods described herein, or any functional variant or fragment thereof. An example of an Icd protein includes, but is not limited to, Icd protein from *E. coli* (Uniprot #P08200). In some embodiments, more than one Icd protein is used in a method described herein. In some embodiments, 2, 3, 4, or 5 different Icd proteins are used in a method described herein.

(28) In some embodiments, the Icd protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of Icd from *E. coli* encoded by the nucleic acid sequence of SEQ ID NO: 2. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of Icd protein from *E. coli* encoded by the nucleic acid sequence of SEQ ID NO: 2. In some embodiments, the Icd protein comprises an amino acid sequence encoded by nucleic acids in SEQ ID NO: 2.

(29) In some embodiments, the Icd protein comprises an amino acid sequence that is at least 75% identical to the amino acid sequence of Icd from *E. coli* provided in SEQ ID NO: 4. In some embodiments, the Icd protein comprises an amino acid sequence that is at least 80%, at least 85%,

at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identical to the amino acid sequence of Icd protein from *E. coli* provided in SEQ ID NO: 4. In some embodiments, the Icd protein comprises the amino acid sequence of SEQ ID NO: 4.

(30) Lactones

(31) According to methods described herein, hydroxy fatty acids may be converted to lactones. According to the present disclosure, the methods provided herein convert subterminal hydroxy fatty acids into γ -lactones, δ -lactones, and combinations thereof. Examples of lactones include, but are not limited to, δ -hexalactone, γ -hexalactone, δ -octalactone, γ -valerolactone, δ -heptalactone, and γ -heptalactone.

(32) In some embodiments, subterminal hydroxy lauric acid is converted to δ -hexalactone, γ -hexalactone, and/or δ -octalactone. In some embodiments, hydroxy tridecanoic acid is converted to γ -valerolactone, δ -heptalactone, and/or γ -heptalactone.

(33) Lactones described herein, in some embodiments, comprise 5 carbons to 8 carbons. In some embodiments, the lactones comprise 5 carbons. In some embodiments, the lactones comprise 6 carbons. In some embodiments, the lactones comprise 7 carbons. In some embodiments, the lactones comprise 8 carbons.

(34) Any lactones produced according to methods described herein may be isolated or purified from a cell culture using any method known in the art. In some embodiments, lactones are isolated or purified from the cell culture with chromatography. In some embodiments, lactones are isolated or purified from the cell culture with organic solvent. In some embodiments, the isolated or purified lactones can be incorporated into a product, e.g., a food, a beverage, a perfume or a cosmetic.

(35) Fatty Acids

(36) According to methods described herein, fatty acids may be converted to hydroxy fatty acids. Any fatty acid may be converted to its hydroxylated form in accordance with methods disclosed herein. Examples of fatty acids include, but are not limited to, pentanoic acid, hexanoic acid, heptanoic acid, octanoic acid, nonanoic acid, decanoic acid, undecanoic acid, dodecanoic acid, tridecanoic acid, tetradecanoic acid, pentadecanoic acid, hexadecanoic acid, heptadecanoic acid, octadecanoic acid, nonadecanoic acid, eicosanoic acid, heneicosanoic acid, docosanoic acid, tricosanoic acid, tetracosanoic acid, pentacosanoic acid, hexacosanoic acid, heptacosanoic acid, octacosanoic acid, nonacosanoic acid, triacontanoic acid, henatriacontanoic acid, dotriacontanoic acid, tritriacontanoic acid, tetratriacontanoic acid, pentatriacontanoic acid, hexatriacontanoic acid, heptatriacontanoic acid, octatriacontanoic acid, nonatriacontanoic acid, and tetracontanoic acid.

Table 1 provides common names, structural formulas, and lipid numbers of exemplary fatty acids.

(37) Fatty acids described herein, in some embodiments, comprise 5 carbons to 40 carbons. In some embodiments, the fatty acids comprise 5 carbons, 6 carbons, 7 carbons, 8 carbons, 9 carbons, 10 carbons, 11 carbons, 12 carbons, 13 carbons, 14 carbons, 15 carbons, 16 carbons, 17 carbons, 18 carbons, 19 carbons, 20 carbons, 21 carbons, 22 carbons, 23 carbons, 24 carbons, 25 carbons, 26 carbons, 27 carbons, 28 carbons, 29 carbons, 30 carbons, 31 carbons, 32 carbons, 33 carbons, 34 carbons, 35 carbons, 36 carbons, 37 carbons, 38 carbons, 39 carbons, 40 carbons, or more carbons.

(38) Fatty acids described herein, in some embodiments, are provided in a medium. In some embodiments, the medium comprising fatty acids comprises a cell culture medium. In some embodiments, the medium comprising fatty acids comprises salts. Salts are combinations of anions and cations. Non-limiting examples of cations include lithium, sodium, potassium magnesium, calcium, and ammonium. Non-limiting examples of anions include acetate, chloride, sulfate, and phosphate. In some embodiments, the medium comprising fatty acids comprises isocitric acid trisodium salt.

(39) In some embodiments, the medium comprising fatty acids comprises a buffer. Examples of buffers include, without limitation, phosphate buffer, Tris buffer, MOPS buffer, HEPES buffer, citrate buffer, acetate buffer, malate buffer, MES buffer, histidine buffer, PIPES buffer, bis-tris

buffer, and ethanolamine buffer.

(40) In some embodiments, the medium comprising fatty acids comprises a detergent. Non-limiting examples of detergents include polysorbate 20 (TWEEN® 20), polyoxyethylenesorbitan monopalmitate (TWEEN® 40), 4-(1,1,3,3-Tetramethylbutyl)phenyl-polyethylene glycol (TRITON™ X-100), sodium dodecyl sulfate (SDS), ethyl trimethylammonium bromide (ETMAB), lauryl trimethyl ammonium bromide (LTAB), and lauryl trimethylammonium chloride (LTAC).

(41) A medium comprising fatty acids may comprise any concentration of fatty acids. In some embodiments, the medium comprising fatty acids comprises 0.5 g/L to 100 g/L fatty acids, e.g., 1 g/L to 10 g/L fatty acids. In some embodiments, the medium comprising fatty acids comprises 0.5 g/L, 1 g/L, 2 g/L, 3 g/L, 4 g/L, 5 g/L, 6 g/L, 7 g/L, 8 g/L, 9 g/L, 10 g/L, 20 g/L, 30 g/L, 40 g/L, 50 g/L, 60 g/L, 70 g/L, 80 g/L, 90 g/L or 100 g/L fatty acids.

(42) Any fatty acid described herein may be hydroxylated according to methods described herein. In some embodiments, pentanoic acid is converted to hydroxylated pentanoic acid, hexanoic acid is converted to hydroxylated hexanoic acid, heptanoic acid is converted to hydroxylated heptanoic acid, octanoic acid is converted to hydroxylated octanoic acid, nonanoic acid is converted to hydroxylated nonanoic acid, decanoic acid is converted to hydroxylated decanoic acid, undecanoic acid is converted to hydroxylated undecanoic acid, dodecanoic acid is converted to hydroxylated dodecanoic acid, tridecanoic acid is converted to hydroxylated tridecanoic acid, tetradecanoic acid is converted to hydroxylated tetradecanoic acid, pentadecanoic acid is converted to hydroxylated pentadecanoic acid, hexadecanoic acid is converted to hydroxylated hexadecanoic acid, heptadecanoic acid is converted to hydroxylated heptadecanoic acid, octadecanoic acid is converted to hydroxylated octadecanoic acid, nonadecanoic acid is converted to hydroxylated nonadecanoic acid, eicosanoic acid is converted to hydroxylated eicosanoic acid, heneicosanoic acid is converted to hydroxylated heneicosanoic acid, docosanoic acid is converted to hydroxylated docosanoic acid, tricosanoic acid is converted to hydroxylated tricosanoic acid, tetracosanoic acid is converted to hydroxylated tetracosanoic acid, pentacosanoic acid is converted to hydroxylated pentacosanoic acid, hexacosanoic acid is converted to hydroxylated hexacosanoic acid, heptacosanoic acid is converted to hydroxylated heptacosanoic acid, octacosanoic acid is converted to hydroxylated octacosanoic acid, nonacosanoic acid is converted to hydroxylated nonacosanoic acid, triacontanoic acid is converted to hydroxylated triacontanoic acid, henatriacontanoic acid is converted to hydroxylated henatriacontanoic acid, dotriacontanoic acid is converted to hydroxylated dotriacontanoic acid, tritriacontanoic acid is converted to hydroxylated tritriacontanoic acid, tetratriacontanoic acid is converted to hydroxylated tetratriacontanoic acid, pentatriacontanoic acid is converted to hydroxylated pentatriacontanoic acid, hexatriacontanoic acid is converted to hydroxylated hexatriacontanoic acid, heptatriacontanoic acid is converted to hydroxylated heptatriacontanoic acid, octatriacontanoic acid is converted to hydroxylated octatriacontanoic acid, nonatriacontanoic acid is converted to hydroxylated nonatriacontanoic acid, and tetracontanoic acid is converted to hydroxylated tetracontanoic acid.

(43) Hydroxy fatty acids, in some embodiments, comprise a hydroxyl group at position ω -1, ω -2, ω -3, or combinations thereof. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -1, ω -2, and ω -3. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -1. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -2. In some embodiments, the hydroxyl fatty acid comprises a hydroxyl group at position ω -3.

(44) In some embodiments, hydroxy fatty acids are contacted with a cell culture. In some embodiments, contacting the hydroxy fatty acids with a cell culture comprises exposing a hydroxy fatty acid with a cell culture for a suitable period sufficient for formation of lactones in the cell culture, if any. In some embodiments, the hydroxy fatty acids are isolated prior to contact with the cell culture. In some embodiments, the hydroxy fatty acids in a medium are contacted with the cell culture.

(45) Cellular Systems

(46) A cellular system refers to any cell or cells that provide for expression of ectopic proteins or heterologous proteins (e.g., CYP450). A cellular system includes, but is not limited to, bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cellular system comprises bacterial cells, yeast cells, or a combination thereof. In some embodiments, the cellular system comprises prokaryotic cells, eukaryotic cells, and combinations thereof. In some embodiments, the cellular system comprises in vitro expression of proteins based on cellular components, such as ribosomes.

(47) Bacterial cells of the present disclosure include, without limitation, *Escherichia* spp., *Streptomyces* spp., *Zymomonas* spp., *Acetobacter* spp., *Citrobacter* spp., *Synechocystis* spp., *Rhizobium* spp., *Clostridium* spp., *Corynebacterium* spp., *Streptococcus* spp., *Xanthomonas* spp., *Lactobacillus* spp., *Lactococcus* spp., *Bacillus* spp., *Alcaligenes* spp., *Pseudomonas* spp., *Aeromonas* spp., *Azotobacter* spp., *Comamonas* spp., *Mycobacterium* spp., *Rhodococcus* spp., *Gluconobacter* spp., *Ralstonia* spp., *Acidithiobacillus* spp., *Microthrix* spp., *Geobacter* spp., *Geobacillus* spp., *Arthrobacter* spp., *Flavobacterium* spp., *Serratia* spp., *Saccharopolyspora* spp., *Thermus* spp., *Stenotrophomonas* spp., *Chromobacterium* spp., *Sinorhizobium* spp., *Saccharopolyspora* spp., *Agrobacterium* spp., *Pantoea* spp., and *Vibrio natriegens*.

(48) Yeast cells of the present disclosure include, without limitation, engineered *Saccharomyces* spp., *Schizosaccharomyces*, *Hansenula*, *Candida*, *Kluyveromyces*, *Yarrowia*, *Candida boidinii*, and *Pichia*. According to the current disclosure, a yeast as claimed herein are eukaryotic, single-celled microorganisms classified as members of the fungus kingdom. Yeasts are unicellular organisms which evolved from multicellular ancestors but with some species useful for the current disclosure being those that have the ability to develop multicellular characteristics by forming strings of connected budding cells known as pseudo hyphae or false hyphae.

(49) In some embodiments, cell pellets are harvested from the cellular system expressing CYP450. In some embodiments, the cell pellets may be resuspended at various concentrations. In some embodiments, the cell pellets are resuspended at a concentration of 1 g/L to 250 g/L. In some embodiments, the cell pellets harvested from the cellular system are resuspended at a concentration of 1 g/L, 10 g/L, 25 g/L, 50 g/L, 75 g/L, 100 g/L, 125 g/L, 150 g/L, 175 g/L, 200 g/L, 225 g/L, or 250 g/L.

(50) Cell Culture

(51) A cell culture refers to any cell or cells that are in a culture. Culturing is the process in which cells are grown under controlled conditions, typically outside of their natural environment. For example, cells, such as yeast cells, may be grown as a cell suspension in liquid nutrient broth. A cell culture includes, but is not limited to, a bacterial cell culture, a yeast cell culture, a plant cell culture, and an animal cell culture. In some embodiments, the cell culture comprises bacterial cells, yeast cells, or a combination thereof.

(52) A bacterial cell culture of the present disclosure comprises bacterial cells including, but not limited to, *Escherichia* spp., *Streptomyces* spp., *Zymomonas* spp., *Acetobacter* spp., *Citrobacter* spp., *Synechocystis* spp., *Rhizobium* spp., *Clostridium* spp., *Corynebacterium* spp., *Streptococcus* spp., *Xanthomonas* spp., *Lactobacillus* spp., *Lactococcus* spp., *Bacillus* spp., *Alcaligenes* spp., *Pseudomonas* spp., *Aeromonas* spp., *Azotobacter* spp., *Comamonas* spp., *Mycobacterium* spp., *Rhodococcus* spp., *Gluconobacter* spp., *Ralstonia* spp., *Acidithiobacillus* spp., *Microthrix* spp., *Geobacter* spp., *Geobacillus* spp., *Arthrobacter* spp., *Flavobacterium* spp., *Serratia* spp., *Saccharopolyspora* spp., *Thermus* spp., *Stenotrophomonas* spp., *Chromobacterium* spp., *Sinorhizobium* spp., *Saccharopolyspora* spp., *Agrobacterium* spp., *Pantoea* spp., and *Vibrio natriegens*.

(53) A yeast cell culture of the present disclosure comprises yeast cells including, but not limited to *Saccharomyces* spp., *Schizosaccharomyces*, *Hansenula*, *Candida*, *Kluyveromyces*, *Yarrowia*, *Candida boidinii*, and *Pichia*.

(54) In some embodiments, a cell culture as described herein is contacted with hydroxy fatty acids as described herein. In some embodiments, a cell culture as described herein is contacted with hydroxy fatty acids as described herein and corn steep liquor, e.g., at a ratio of 1:1 hydroxy fatty acids to corn steep liquor.

(55) In some embodiments, cells are cultured at a temperature of 16° C. to 40° C. For example, cells may be cultured at a temperature of 16° C., 17° C., 18° C., 19° C., 20° C., 21° C., 22° C., 23° C., 24° C., 25° C., 26° C., 27° C., 28° C., 29° C., 30° C., 31° C., 32° C., 33° C., 34° C., 35° C., 36° C., 37° C., 38° C., 39° C. or 40° C.

(56) In some embodiments, cells are cultured for a period of time of 0.5 hours to 96 hours, or more. For example, cells may be cultured for a period of time of 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, or 72 hours. Typically, cells, such as bacterial cells, are cultured for a period of time of 12 to 24 hours. In some embodiments, cells are cultured for 12 to 24 hours at a temperature of 37° C. In some embodiments, cells are cultured for 12 to 24 hours at a temperature of 16° C.

(57) In some embodiments, cells are cultured to a density of $1 \times 10^{0.8}$ ($OD_{600} < 1$) to $2 \times 10^{1.1}$ ($OD \sim 200$) viable cells/ml cell culture medium. In some embodiments, cells are cultured to a density of $1 \times 10^{0.8}$, $2 \times 10^{0.8}$, $3 \times 10^{0.8}$, $4 \times 10^{0.8}$, $5 \times 10^{0.8}$, $6 \times 10^{0.8}$, $7 \times 10^{0.8}$, $8 \times 10^{0.8}$, $9 \times 10^{0.8}$, $1 \times 10^{0.9}$, $2 \times 10^{0.9}$, $3 \times 10^{0.9}$, $4 \times 10^{0.9}$, $5 \times 10^{0.9}$, $6 \times 10^{0.9}$, $7 \times 10^{0.9}$, $8 \times 10^{0.9}$, $9 \times 10^{0.9}$, $1 \times 10^{1.0}$, $2 \times 10^{1.0}$, $3 \times 10^{1.0}$, $4 \times 10^{1.0}$, $5 \times 10^{1.0}$, $6 \times 10^{1.0}$, $7 \times 10^{1.0}$, $8 \times 10^{1.0}$, $9 \times 10^{1.0}$, $1 \times 10^{1.1}$, or $2 \times 10^{1.1}$ viable cells/ml. (Conversion factor: $OD\ 1 = 8 \times 10^{0.8}$ cells/ml).

(58) Lactones

(59) Lactones are cyclic carboxylic esters containing a 1-oxacycloalkane-2-one structure ($-(C=O)-O-$) or analogues formed from intramolecular esterification. The present invention relates to lactones (for example, γ -lactones and δ -lactones) formed from straight chain hydroxy carboxylic acids in which the hydroxyl group has reacted with the acid group to form a cyclic carboxylic ester.

(60) The International Union of Pure and Applied Chemistry (IUPAC) names of lactones are derived by adding the term “lactone” at the end of the name of the parent acid. The common names of lactones, used more often than IUPAC names, are formed by changing the -ic acid ending by the hydroxy acid to -olactone. Greek letter designates the carbon atom that bears the hydroxy group to close the ring in the parent hydroxyl carboxylic acid. In other words, Greek letter prefix specifies the number of carbon atoms in the heterocycle—that is, the distance between the relevant $-OH$ and the $-COOH$ groups along said backbone of the parent hydroxyl carboxylic acid. The first carbon atom after the carbon in the $-COOH$ group on the parent compound is labelled α , the second will be labeled β , and so forth. Therefore, the Greek letter prefixes also indicate the size of the lactone ring: α -lactone=3-membered ring, β -lactone=4-membered ring, γ -lactone=5-membered ring; and δ -lactone=6-membered ring.

(61) To illustrate, the lactone derived from the cyclization of γ -hydroxybutyric acid is known as 4-hydroxybutanoic acid lactone or γ -butyrolactone. Similarly, 5-hydroxypentanoic acid is referred as δ -hydroxypentanoic acid and the corresponding lactone is known as 5-hydroxypentanoic acid lactone or δ -valerolactone.

(62) Hydroxy Straight Chain Fatty Acids and Subterminal Hydroxy Groups

(63) Straight chain fatty acid consists of a straight hydrocarbon chain that includes, at the end, a carboxyl group ($-COOH$). It is the presence of this carboxylic acid group that makes this chemical entity an acid. The straight chain fatty acids may optionally have unsaturated bond between the carbon atom. Branched chain fatty acids are usually saturated fatty acids with one or more alkyl branches substituted on the carbon chain.

(64) One or more hydrogen atoms attached to the hydrocarbon back bone of the straight chain fatty acid may also be replaced by a hydroxyl ($-OH$) group. Straight chain carboxylic acid with one or more hydroxyl groups is referred as a hydroxy carboxylic acid. Ricinoleic acid is a naturally occurring hydroxy carboxylic acid.

(65) The present invention is related to the replacement or substitution of a hydrogen atom in a saturated straight chain fatty acid with a hydroxyl group using a recombinant microbial organism expressing recombinant protein with the hydroxylase enzyme activity. In a preferred embodiment, substitution of the hydroxyl group takes place at one of the —CH₂— groups closest to the terminal methyl end group of the straight chain fatty acid. In general terminology, n or ω refers to the methyl end group of the straight chain fatty acid while the number n-x or ω-x refers to the location of the carbon atom on the back bone where x indicates the number of carbons away from the methyl end group. To illustrate, in the example of stearic acid (or octadecanoic acid), a C₁₈:0 straight chain fatty acid with the formula CH₃(CH₂)₁₆COOH, ω-3 would mean the fifteenth carbon in the back bone which is three carbon removed from the methyl end group. Similarly, in the same example, ω-1 would mean the seventeenth carbon in the back bone which is one carbon removed from the methyl end group.

(66) As used in the present invention, when a hydroxyl group is introduced into a straight chain fatty acid specifically at positions ω-1, ω-2 and ω-3 using the recombinant DNA technology according to the present invention, the resulting hydroxylated straight chain fatty acid molecule is referred to as a subterminal hydroxy fatty acid, and the hydroxyl group at positions ω-1, ω-2 and ω-3 in such fatty acid molecule are referred to as subterminal hydroxyl groups.

(67) L1

(68) Microbial Culture Media

(69) The microbial organisms useful in the present invention including wild type and recombinant bacterial, yeast and fungal strains are grown in specified culture medium. The microbial culture medium may be minimal growth medium or rich growth medium. The minimal growth medium comprises a fortified mixture of inorganic compounds and one or more organic nutrients such as glucose, sucrose, or glycerol. The rich growth medium comprises yeast extract, peptone and organic nutrient such as dextrose, sucrose or glycerol. In preferred embodiments, the rich growth medium is supplemented with corn steep liquor to enhance the growth of the microbial organisms.

(70) Synthetic Biology

(71) Standard recombinant DNA and molecular cloning techniques used here are well known in the art and are described, for example, by Sambrook, J., Fritsch, E. F. and Maniatis, T. MOLECULAR CLONING: A LABORATORY MANUAL, 2nd ed.; Cold Spring Harbor Laboratory: Cold Spring Harbor, N.Y., 1989 (hereinafter “Maniatis”); and by Silhavy, T. J., Bennis, M. L. and Enquist, L. W. EXPERIMENTS WITH GENE FUSIONS; Cold Spring Harbor Laboratory: Cold Spring Harbor, N.Y., 1984; and Ausubel, F. M. et al., IN CURRENT PROTOCOLS IN MOLECULAR BIOLOGY, published by GREENE PUBLISHING AND WILEY-INTERSCIENCE, 1987; (the entirety of each of which is hereby incorporated herein by reference).

(72) Bacterial Production Systems

(73) Expression of proteins in prokaryotes is most often carried out in a bacterial host cell with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins. Fusion vectors add a number of amino acids to a protein encoded therein, usually to the amino terminus of the recombinant protein. Such fusion vectors typically serve three purposes: 1) to increase expression of recombinant protein; 2) to increase the solubility of the recombinant protein; and 3) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such vectors are within the scope of the present disclosure.

(74) In an embodiment, the expression vector includes those genetic elements for expression of the recombinant polypeptide in bacterial cells. The elements for transcription and translation in the bacterial cell can include a promoter, a coding region for the protein complex, and a transcriptional terminator.

(75) A person of ordinary skill in the art will be aware of the molecular biology techniques available for the preparation of expression vectors. The polynucleotide used for incorporation into the expression vector of the subject technology, as described above, can be prepared by routine techniques such as polymerase chain reaction (PCR).

(76) A number of molecular biology techniques have been developed to operably link DNA to vectors via complementary cohesive termini. In one embodiment, complementary homopolymer tracts can be added to the nucleic acid molecule to be inserted into the vector DNA. The vector and nucleic acid molecule are then joined by hydrogen bonding between the complementary homopolymeric tails to form recombinant DNA molecules.

(77) In an alternative embodiment, synthetic linkers containing one or more restriction sites are used to operably link the polynucleotide of the subject technology to the expression vector. In an embodiment, the polynucleotide is generated by restriction endonuclease digestion. In an embodiment, the nucleic acid molecule is treated with bacteriophage T4 DNA polymerase or *E. coli* DNA polymerase I, enzymes that remove protruding, 3'-single-stranded termini with their 3'-5'-exonucleolytic activities and fill-in recessed 3'-ends with their polymerizing activities, thereby generating blunt-ended DNA segments. The blunt-ended segments are then incubated with a large molar excess of linker molecules in the presence of an enzyme that is able to catalyze the ligation of blunt-ended DNA molecules, such as bacteriophage T4 DNA ligase. Thus, the product of the reaction is a polynucleotide carrying polymeric linker sequences at its ends. These polynucleotides are then cleaved with the appropriate restriction enzyme and ligated to an expression vector that has been cleaved with an enzyme that produces termini compatible with those of the polynucleotide.

(78) Alternatively, a vector having ligation-independent cloning (LIC) sites can be employed. The required PCR amplified polynucleotide can then be cloned into the LIC vector without restriction digest or ligation (Aslanidis and de Jong, NUCL. ACID. RES. 18 6069-74, (1990), Haun, et al, BIOTECHNIQUES 13, 515-18 (1992), which is incorporated herein by reference to the extent it is consistent herewith).

(79) In an embodiment, in order to isolate and/or modify the polynucleotide of interest for insertion into the chosen plasmid, it is suitable to use PCR. Appropriate primers for use in PCR preparation of the sequence can be designed to isolate the required coding region of the nucleic acid molecule, to add restriction endonuclease or LIC sites and to place the coding region in the desired reading frame.

(80) In an embodiment, a polynucleotide for incorporation into an expression vector of the subject technology is prepared by the use of PCR using appropriate oligonucleotide primers. The coding region is amplified, whilst the primers themselves become incorporated into the amplified sequence product. In an embodiment, the amplification primers contain restriction endonuclease recognition sites, which allow the amplified sequence product to be cloned into an appropriate vector.

(81) The expression vectors can be introduced into plant or microbial host cells by conventional transformation or transfection techniques. Transformation of appropriate cells with an expression vector of the subject technology is accomplished by methods known in the art and typically depends on both the type of vector and cell. Suitable techniques include calcium phosphate or calcium chloride co-precipitation, DEAE-dextran mediated transfection, lipofection, chemoporation or electroporation.

(82) Successfully transformed cells, that is, those cells containing the expression vector, can be identified by techniques well known in the art. For example, cells transfected with an expression vector of the subject technology can be cultured to produce polypeptides described herein. Cells can be examined for the presence of the expression vector DNA by techniques well known in the art.

(83) The host cells can contain a single copy of the expression vector described previously, or alternatively, multiple copies of the expression vector.

(84) In some embodiments, the transformed cell is an animal cell, an insect cell, a plant cell, an algal cell, a fungal cell, or a yeast cell. In some embodiments, the cell is a plant cell selected from the group consisting of: canola plant cell, a rapeseed plant cell, a palm plant cell, a sunflower plant cell, a cotton plant cell, a corn plant cell, a peanut plant cell, a flax plant cell, a sesame plant cell, a soybean plant cell, and a *petunia* plant cell.

(85) Microbial host cell expression systems and expression vectors containing regulatory sequences that direct high-level expression of foreign proteins are well known to those skilled in the art. Any of these could be used to construct vectors for expression of the recombinant polypeptide of the subject technology in a microbial host cell. These vectors could then be introduced into appropriate microorganisms via transformation to allow for high level expression of the recombinant polypeptide of the subject technology.

(86) Vectors or cassettes useful for the transformation of suitable microbial host cells are well known in the art. Typically the vector or cassette contains sequences directing transcription and translation of the relevant polynucleotide, a selectable marker, and sequences allowing autonomous replication or chromosomal integration. Suitable vectors comprise a region 5' of the polynucleotide which harbors transcriptional initiation controls and a region 3' of the DNA fragment which controls transcriptional termination. It is preferred for both control regions to be derived from genes homologous to the transformed host cell, although it is to be understood that such control regions need not be derived from the genes native to the specific species chosen as a host.

(87) Initiation control regions or promoters, which are useful to drive expression of the recombinant polypeptide in the desired microbial host cell are numerous and familiar to those skilled in the art. Virtually any promoter capable of driving these genes is suitable for the subject technology including but not limited to CYCI, HIS3, GALI, GALIO, ADHI, PGK, PH05, GAPDH, ADCI, TRPI, URA3, LEU2, ENO, TPI (useful for expression in *Saccharomyces*); AOXI (useful for expression in *Pichia*); and lac, trp, JPL, IPR, T7, tac, and trc (useful for expression in *Escherichia coli*).

(88) Termination control regions may also be derived from various genes native to the microbial hosts. A termination site optionally may be included for the microbial hosts described herein.

(89) In plant cells, the expression vectors of the subject technology can include a coding region operably linked to promoters capable of directing expression of the recombinant polypeptide of the subject technology in the desired tissues at the desired stage of development. For reasons of convenience, the polynucleotides to be expressed may comprise promoter sequences and translation leader sequences derived from the same polynucleotide. 3' non-coding sequences encoding transcription termination signals should also be present. The expression vectors may also comprise one or more introns in order to facilitate polynucleotide expression.

(90) For plant host cells, any combination of any promoter and any terminator capable of inducing expression of a coding region may be used in the vector sequences of the subject technology. Some suitable examples of promoters and terminators include those from nopaline synthase (nos), octopine synthase (ocs) and cauliflower mosaic virus (CaMV) genes. One type of efficient plant promoter that may be used is a high-level plant promoter. Such promoters, in operable linkage with an expression vector of the subject technology should be capable of promoting the expression of the vector. High level plant promoters that may be used in the subject technology include the promoter of the small subunit(s) of the ribulose-1,5-bisphosphate carboxylase for example from soybean (Berry-Lowe et al., J. MOLECULAR AND APP. GEN., 1:483-98 (1982), the entirety of which is hereby incorporated herein to the extent it is consistent herewith), and the promoter of the chlorophyll binding protein. These two promoters are known to be light-induced in plant cells (see, for example, GENETIC ENGINEERING OF PLANTS, AN AGRICULTURAL PERSPECTIVE, A. Cashmore, Plenum, N.Y. (1983), pages 29-38; Coruzzi, G. et al., THE JOURNAL OF BIOLOGICAL CHEMISTRY, 258:1399 (1983), and Dunsmuir, P. et al., JOURNAL OF MOLECULAR AND APPLIED GENETICS, 2:285 (1983), each of which is hereby incorporated

herein by reference to the extent they are consistent herewith).

(91) Analysis of Sequence Similarity Using Identity Scoring

(92) As used herein “sequence identity” refers to the extent to which two optimally aligned polynucleotide or peptide sequences are invariant throughout a window of alignment of components, e.g., nucleotides or amino acids. An “identity fraction” for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in reference sequence segment, i.e., the entire reference sequence or a smaller defined part of the reference sequence.

(93) As used herein, the term “percent sequence identity” or “percent identity” refers to the percentage of identical nucleotides in a linear polynucleotide sequence of a reference (“query”) polynucleotide molecule (or its complementary strand) as compared to a test (“subject”) polynucleotide molecule (or its complementary strand) when the two sequences are optimally aligned (with appropriate nucleotide insertions, deletions, or gaps totaling less than 20 percent of the reference sequence over the window of comparison). Optimal alignment of sequences for aligning a comparison window are well known to those skilled in the art and may be conducted by tools such as the local homology algorithm of Smith and Waterman, the homology alignment algorithm of Needleman and Wunsch, the search for similarity method of Pearson and Lipman, and preferably by computerized implementations of these algorithms such as GAP, BESTFIT, FASTA, and TFASTA available as part of the GCG® Wisconsin Package® (Accelrys Inc., Burlington, MA). An “identity fraction” for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in the reference sequence segment, i.e., the entire reference sequence or a smaller defined part of the reference sequence. Percent sequence identity is represented as the identity fraction multiplied by 100. The comparison of one or more polynucleotide sequences may be to a full-length polynucleotide sequence or a portion thereof, or to a longer polynucleotide sequence. For purposes of this disclosure “percent identity” may also be determined using BLASTX version 2.0 for translated nucleotide sequences and BLASTN version 2.0 for polynucleotide sequences.

(94) The percent of sequence identity is preferably determined using the “Best Fit” or “Gap” program of the Sequence Analysis Software Package™ (Version 10; Genetics Computer Group, Inc., Madison, WI). “Gap” utilizes the algorithm of Needleman and Wunsch (Needleman and Wunsch, JOURNAL OF MOLECULAR BIOLOGY 48:443-53, 1970) to find the alignment of two sequences that maximizes the number of matches and minimizes the number of gaps. “BestFit” performs an optimal alignment of the best segment of similarity between two sequences and inserts gaps to maximize the number of matches using the local homology algorithm of Smith and Waterman (Smith and Waterman, ADVANCES IN APPLIED MATHEMATICS, 2:482-489, 1981, Smith et al., NUCLEIC ACIDS RESEARCH 11:2205-2220, 1983). The percent identity is most preferably determined using the “Best Fit” program.

(95) Useful methods for determining sequence identity are also disclosed in the Basic Local Alignment Search Tool (BLAST) programs which are publicly available from National Center Biotechnology Information (NCBI) at the National Library of Medicine, National Institute of Health, Bethesda, Md. 20894; see BLAST Manual, Altschul et al., NCBI, NLM, NIH; Altschul et al., J. MOL. BIOL. 215:403-10 (1990); version 2.0 or higher of BLAST programs allows the introduction of gaps (deletions and insertions) into alignments; for peptide sequence BLASTX can be used to determine sequence identity; and, for polynucleotide sequence BLASTN can be used to determine sequence identity.

(96) As used herein, the term “substantial percent sequence identity” refers to a percent sequence identity of at least about 70% sequence identity, at least about 80% sequence identity, at least about 85% identity, at least about 90% sequence identity, or even greater sequence identity, such as about 98% or about 99% sequence identity. Thus, one embodiment of the disclosure is a polynucleotide

molecule that has at least about 70% sequence identity, at least about 80% sequence identity, at least about 85% identity, at least about 90% sequence identity, or even greater sequence identity, such as about 98% or about 99% sequence identity with a polynucleotide sequence described herein. Polynucleotide molecules that have the activity of the Cytochrome P450 genes of the current disclosure are capable of directing the production of a variety of γ - and delta-lactones. Such polynucleotide molecules can have a substantial percent sequence identity to the polynucleotide sequences provided herein and are encompassed within the scope of this disclosure.

(97) Identity and Similarity

(98) Identity is the fraction of amino acids that are the same between a pair of sequences after an alignment of the sequences (which can be done using only sequence information or structural information or some other information, but usually it is based on sequence information alone), and similarity is the score assigned based on an alignment using some similarity matrix. The similarity index can be any one of the following BLOSUM62, PAM250, or GONNET, or any matrix used by one skilled in the art for the sequence alignment of proteins.

(99) Identity is the degree of correspondence between two sub-sequences (no gaps between the sequences). An identity of 25% or higher implies similarity of function, while 18-25% implies similarity of structure or function. Keep in mind that two completely unrelated or random sequences (that are greater than 100 residues) can have higher than 20% identity. Similarity is the degree of resemblance between two sequences when they are compared. This is dependent on their identity.

Explanation of Terms Used Herein

(100) “Coding sequence” is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to a DNA sequence that encodes for a specific amino acid sequence. “Protein Expression”. Protein production can occur after gene expression. It consists of the stages after DNA has been transcribed to messenger RNA (mRNA). The mRNA is then translated into polypeptide chains, which are ultimately folded into proteins. DNA is present in the cells through transfection—a process of deliberately introducing nucleic acids into cells. The term is often used for non-viral methods in eukaryotic cells. It may also refer to other methods and cell types, although other terms are preferred: “transformation” is more often used to describe non-viral DNA transfer in bacteria, non-animal eukaryotic cells, including plant cells. In animal cells, transfection is the preferred term as transformation is also used to refer to progression to a cancerous state (carcinogenesis) in these cells. Transduction is often used to describe virus-mediated DNA transfer. Transformation, transduction, and viral infection are included under the definition of transfection for this application.

(101) As used herein, the singular forms “a, an” and “the” include plural references unless the content clearly dictates otherwise.

(102) To the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

(103) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

(104) The term “complementary” is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is used without limitation to describe the relationship between nucleotide bases that are capable to hybridizing to one another. For example, with respect to DNA, adenosine is complementary to thymine and cytosine is complementary to guanine. Accordingly, the subjection technology also includes isolated nucleic acid fragments that are complementary to the complete sequences as reported in the accompanying Sequence Listing as well as those substantially similar nucleic acid sequences.

(105) The terms “nucleic acid” and “nucleotide” are to be given their respective ordinary and

customary meanings to a person of ordinary skill in the art and are used without limitation to refer to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogues of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally-occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified or degenerate variants thereof (e.g., degenerate codon substitutions) and complementary sequences, as well as the sequence explicitly indicated.

(106) The term “isolated” is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and when used in the context of an isolated nucleic acid or an isolated polypeptide, is used without limitation to refer to a nucleic acid or polypeptide that, by the hand of man, exists apart from its native environment and is therefore not a product of nature. An isolated nucleic acid or polypeptide can exist in a purified form or can exist in a non-native environment such as, for example, in a transgenic host cell.

(107) The term “degenerate variant” refers to a nucleic acid sequence having a residue sequence that differs from a reference nucleic acid sequence by one or more degenerate codon substitutions. Degenerate codon substitutions can be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed base and/or deoxyinosine residues. A nucleic acid sequence and all of its degenerate variants will express the same amino acid or polypeptide.

(108) The terms “polypeptide,” “protein,” and “peptide” are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art; the three terms are sometimes used interchangeably, and are used without limitation to refer to a polymer of amino acids, or amino acid analogs, regardless of its size or function. Although “protein” is often used in reference to relatively large polypeptides, and “peptide” is often used in reference to small polypeptides, usage of these terms in the art overlaps and varies. The term “polypeptide” as used herein refers to peptides, polypeptides, and proteins, unless otherwise noted. The terms “protein,” “polypeptide,” and “peptide” are used interchangeably herein when referring to a polynucleotide product. Thus, exemplary polypeptides include polynucleotide products, naturally occurring proteins, homologs, orthologs, paralogs, fragments and other equivalents, variants, and analogs of the foregoing.

(109) The terms “polypeptide fragment” and “fragment,” when used in reference to a reference polypeptide, are to be given their ordinary and customary meanings to a person of ordinary skill in the art and are used without limitation to refer to a polypeptide in which amino acid residues are deleted as compared to the reference polypeptide itself, but where the remaining amino acid sequence is usually identical to the corresponding positions in the reference polypeptide. Such deletions can occur at the amino-terminus or carboxy-terminus of the reference polypeptide, or alternatively both.

(110) The term “functional fragment” of a polypeptide or protein refers to a peptide fragment that is a portion of the full-length polypeptide or protein, and has substantially the same biological activity, or carries out substantially the same function as the full-length polypeptide or protein (e.g., carrying out the same enzymatic reaction).

(111) The terms “variant polypeptide,” “modified amino acid sequence” or “modified polypeptide,” which are used interchangeably, refer to an amino acid sequence that is different from the reference polypeptide by one or more amino acids, e.g., by one or more amino acid substitutions, deletions, and/or additions. In an aspect, a variant is a “functional variant” which retains some or all of the ability of the reference polypeptide.

(112) The term “functional variant” further includes conservatively substituted variants. The term “conservatively substituted variant” refers to a peptide having an amino acid sequence that differs from a reference peptide by one or more conservative amino acid substitutions and maintains some or all of the activity of the reference peptide. A “conservative amino acid substitution” is a

substitution of an amino acid residue with a functionally similar residue. Examples of conservative substitutions include the substitution of one non-polar (hydrophobic) residue such as isoleucine, valine, leucine or methionine for another; the substitution of one charged or polar (hydrophilic) residue for another such as between arginine and lysine, between glutamine and asparagine, between threonine and serine; the substitution of one basic residue such as lysine or arginine for another; or the substitution of one acidic residue, such as aspartic acid or glutamic acid for another; or the substitution of one aromatic residue, such as phenylalanine, tyrosine, or tryptophan for another. Such substitutions are expected to have little or no effect on the apparent molecular weight or isoelectric point of the protein or polypeptide. The phrase “conservatively substituted variant” also includes peptides wherein a residue is replaced with a chemically-derivatized residue, provided that the resulting peptide maintains some or all of the activity of the reference peptide as described herein.

(113) The term “variant,” in connection with the polypeptides of the subject technology, further includes a functionally active polypeptide having an amino acid sequence at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, and even 100% identical to the amino acid sequence of a reference polypeptide.

(114) The term “homologous” in all its grammatical forms and spelling variations refers to the relationship between polynucleotides or polypeptides that possess a “common evolutionary origin,” including polynucleotides or polypeptides from super families and homologous polynucleotides or proteins from different species (Reeck et al., CELL 50:667, 1987). Such polynucleotides or polypeptides have sequence homology, as reflected by their sequence similarity, whether in terms of percent identity or the presence of specific amino acids or motifs at conserved positions. For example, two homologous polypeptides can have amino acid sequences that are at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, and even 100% identical.

(115) “Suitable regulatory sequences” is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to nucleotide sequences located upstream (5' non-coding sequences), within, or downstream (3' non-coding sequences) of a coding sequence, and which influence the transcription, RNA processing or stability, or translation of the associated coding sequence. Regulatory sequences may include promoters, translation leader sequences, introns, and polyadenylation recognition sequences.

(116) “Promoter” is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to a DNA sequence capable of controlling the expression of a coding sequence or functional RNA. In general, a coding sequence is located 3' to a promoter sequence. Promoters may be derived in their entirety from a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic DNA segments. It is understood by those skilled in the art that different promoters may direct the expression of a gene in different tissues or cell types, or at different stages of development, or in response to different environmental conditions. Promoters, which cause a gene to be expressed in most cell types at most times, are commonly referred to as “constitutive promoters.” It is further recognized that since in most cases the exact boundaries of regulatory sequences have not been completely defined, DNA fragments of different lengths may have identical promoter activity.

(117) The term “operably linked” refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one is affected by the other. For example, a promoter is operably linked with a coding sequence when it is capable of affecting the expression of that coding sequence (i.e., that the coding sequence is under the transcriptional control of the promoter).

Coding sequences can be operably linked to regulatory sequences in sense or antisense orientation. (118) The term “expression” as used herein, is to be given its ordinary and customary meaning to a person of ordinary skill in the art, and is used without limitation to refer to the transcription and stable accumulation of sense (mRNA) or antisense RNA derived from the nucleic acid fragment of the subject technology. “Over-expression” refers to the production of a gene product in transgenic or recombinant organisms that exceeds levels of production in normal or non-transformed organisms.

(119) “Transformation” is to be given its ordinary and customary meaning to a person of reasonable skill in the field, and is used without limitation to refer to the transfer of a polynucleotide into a target cell for further expression by that cell. The transferred polynucleotide can be incorporated into the genome or chromosomal DNA of a target cell, resulting in genetically stable inheritance, or it can replicate independent of the host chromosomal. Host organisms containing the transformed nucleic acid fragments are referred to as “transgenic” or “recombinant” or “transformed” organisms.

(120) The terms “transformed,” “transgenic,” and “recombinant,” when used herein in connection with host cells, are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art, and are used without limitation to refer to a cell of a host organism, such as a plant or microbial cell, into which a heterologous nucleic acid molecule has been introduced. The nucleic acid molecule can be stably integrated into the genome of the host cell, or the nucleic acid molecule can be present as an extrachromosomal molecule. Such an extrachromosomal molecule can be auto-replicating. Transformed cells, tissues, or subjects are understood to encompass not only the end product of a transformation process, but also transgenic progeny thereof.

(121) The terms “recombinant,” “heterologous,” and “exogenous,” when used herein in connection with polynucleotides, are to be given their ordinary and customary meanings to a person of ordinary skill in the art, and are used without limitation to refer to a polynucleotide (e.g., a DNA sequence or a gene) that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous gene in a host cell includes a gene that is endogenous to the particular host cell but has been modified through, for example, the use of site-directed mutagenesis or other recombinant techniques. The terms also include non-naturally occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position or form within the host cell in which the element is not ordinarily found.

(122) Similarly, the terms “recombinant,” “heterologous,” and “exogenous,” when used herein in connection with a polypeptide or amino acid sequence, means a polypeptide or amino acid sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, recombinant DNA segments can be expressed in a host cell to produce a recombinant polypeptide.

(123) The terms “plasmid,” “vector,” and “cassette” are to be given their respective ordinary and customary meanings to a person of ordinary skill in the art and are used without limitation to refer to an extra chromosomal element often carrying genes which are not part of the central metabolism of the cell, and usually in the form of circular double-stranded DNA molecules. Such elements may be autonomously replicating sequences, genome integrating sequences, phage or nucleotide sequences, linear or circular, of a single- or double-stranded DNA or RNA, derived from any source, in which a number of nucleotide sequences have been joined or recombined into a unique construction which is capable of introducing a promoter fragment and DNA sequence for a selected gene product along with appropriate 3' untranslated sequence into a cell. “Transformation cassette” refers to a specific vector containing a foreign gene and having elements in addition to the foreign gene that facilitate transformation of a particular host cell. “Expression cassette” refers to a specific vector containing a foreign gene and having elements in addition to the foreign gene that allow for enhanced expression of that gene in a foreign host.

(124) The term “cellular system” as used herein refers to any cells that provide for the expression of ectopic proteins. In some embodiments, the cellular system comprises bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cellular system, comprises prokaryotic cells and/or eukaryotic cells. In some embodiments, the cellular system comprises in vitro expression of proteins based on cellular components, such as ribosomes.

(125) The term “cell culture” as used herein refers to a composition comprising cells. In some embodiments, the cell culture is a liquid (e.g., a broth culture). In some embodiments, the cell culture is a solid (e.g., a stab culture). In some embodiments, the cell culture comprises bacterial cells, yeast cells, plant cells, and animal cells. In some embodiments, the cell culture comprises prokaryotic cells and/or eukaryotic cells. In some embodiments, the cell culture comprises in vitro expression of proteins based on cellular components, such as ribosomes.

(126) The terms “incubating” and “incubation” as used herein refers to a process of mixing two or more chemical or biological entities (such as a chemical compound and an enzyme) and allowing them to interact under conditions favorable for producing a new chemical entity such as hydroxy fatty acid (e.g., a subterminal hydroxy fatty acid) as described herein or a lactone (e.g., a gamma-lactone, a delta-lactone, or combinations thereof) as described herein.

(127) “Percent (%) amino acid sequence identity” with respect to the variant polypeptide sequences of the subject technology refers to the percentage of amino acid residues in a candidate sequence that are identical with the amino acid residues of a reference polypeptide, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity.

(128) Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN, ALIGN-2 or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full-length of the sequences being compared. For example, the % amino acid sequence identity may be determined using the sequence comparison program NCBI-BLAST2. The NCBI-BLAST2 sequence comparison program may be downloaded from ncbi.nlm.nih.gov. NCBI BLAST2 uses several search parameters, wherein all of those search parameters are set to default values including, for example, unmask yes, strand=all, expected occurrences 10, minimum low complexity length=15/5, multi-pass e-value=0.01, constant for multi-pass=25, dropoff for final gapped alignment=25 and scoring matrix=BLOSUM62. In situations where NCBI-BLAST2 is employed for amino acid sequence comparisons, the % amino acid sequence identity of a given amino acid sequence A to, with, or against a given amino acid sequence B (which can alternatively be phrased as a given amino acid sequence A that has or comprises a certain % amino acid sequence identity to, with, or against a given amino acid sequence B) is calculated as follows: 100 times the fraction X/Y where X is the number of amino acid residues scored as identical matches by the sequence alignment program NCBI-BLAST2 in that program's alignment of A and B, and where Y is the total number of amino acid residues in B. It will be appreciated that where the length of amino acid sequence A is not equal to the length of amino acid sequence B, the % amino acid sequence identity of A to B will not equal the % amino acid sequence identity of B to A.

(129) In this sense, techniques for determining amino acid sequence “similarity” are well known in the art. In general, “similarity” refers to the exact amino acid to amino acid comparison of two or more polypeptides at the appropriate place, where amino acids are identical or possess similar chemical and/or physical properties such as charge or hydrophobicity. A so-termed “percent similarity” may then be determined between the compared polypeptide sequences. Techniques for determining nucleic acid and amino acid sequence identity also are well known in the art and include determining the nucleotide sequence of the mRNA for that gene (usually via a cDNA intermediate) and determining the amino acid sequence encoded therein, and comparing this to a

second amino acid sequence. In general, “identity” refers to an exact nucleotide to nucleotide or amino acid to amino acid correspondence of two polynucleotides or polypeptide sequences, respectively. Two or more polynucleotide sequences can be compared by determining their “percent identity”, as can two or more amino acid sequences. The programs available in the Wisconsin Sequence Analysis Package, Version 8 (available from Genetics Computer Group, Madison, Wis.), for example, the GAP program, are capable of calculating both the identity between two polynucleotides and the identity and similarity between two polypeptide sequences, respectively. Other programs for calculating identity or similarity between sequences are known by those skilled in the art.

(130) An amino acid position “corresponding to” a reference position refers to a position that aligns with a reference sequence, as identified by aligning the amino acid sequences. Such alignments can be done by hand or by using well-known sequence alignment programs such as ClustalW2, Blast 2, etc.

(131) Unless specified otherwise, the percent identity of two polypeptide or polynucleotide sequences refers to the percentage of identical amino acid residues or nucleotides across the entire length of the shorter of the two sequences.

(132) “Transformation” is used according to its ordinary and customary meaning as understood by a person of ordinary skill in the art, and is used without limitation to refer to the transfer of a polynucleotide into a target cell. The transferred polynucleotide can be incorporated into the genome or chromosomal DNA of a target cell, resulting in genetically stable inheritance, or it can replicate independent of the host chromosomal. Host organisms containing the transformed nucleic acid fragments are referred to as “transgenic” or “recombinant” or “transformed” organisms.

(133) The terms “recombinant,” “heterologous,” and “exogenous,” when used herein in connection with polynucleotides, are used according to their ordinary and customary meanings as understood by a person of ordinary skill in the art, and are used without limitation to refer to a polynucleotide (e.g., a DNA sequence or a gene) that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous gene in a host cell includes a gene that is endogenous to the particular host cell but has been modified through, for example, the use of site-directed mutagenesis or other recombinant techniques. The terms also include non-naturally occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position or form within the host cell in which the element is not ordinarily found.

(134) Similarly, the terms “recombinant,” “heterologous,” and “exogenous,” when used herein in connection with a polypeptide or amino acid sequence, means a polypeptide or amino acid sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, recombinant DNA segments can be expressed in a host cell to produce a recombinant polypeptide.

(135) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs. Although any methods and materials similar to or equivalent to those described herein can be used in the practice or testing of the present disclosure, the preferred materials and methods are described below.

(136) The disclosure will be more fully understood upon consideration of the following non-limiting Examples. It should be understood that these Examples, while indicating preferred embodiments of the subject technology, are given by way of illustration only. From the above discussion and these Examples, one skilled in the art can ascertain the essential characteristics of the subject technology, and without departing from the spirit and scope thereof, can make various changes and modifications of the subject technology to adapt it to various uses and conditions.

(137) All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same,

equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features. From the above description, one skilled in the art can easily ascertain the essential characteristics of the present disclosure, and without departing from the spirit and scope thereof, can make various changes and modifications of the present disclosure to adapt it to various usages and conditions. Thus, other embodiments are also within the claims.

EQUIVALENTS AND SCOPE

(138) Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present disclosure, as defined in the following claims.

(139) The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

(140) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

(141) As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

(142) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

(143) It should also be understood that, unless clearly indicated to the contrary, in any methods

claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

(144) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03. It should be appreciated that embodiments described in this document using an open-ended transitional phrase (e.g., “comprising”) are also contemplated, in alternative embodiments, as “consisting of” and “consisting essentially of” the feature described by the open-ended transitional phrase. For example, if the disclosure describes “a composition comprising A and B”, the disclosure also contemplates the alternative embodiments “a composition consisting of A and B” and “a composition consisting essentially of A and B”.

(145) Furthermore, the present disclosure encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should be understood that, in general, where the present disclosure, or aspects of the present disclosure, is/are referred to as comprising particular elements and/or features, certain embodiments of the present disclosure or aspects of the present disclosure consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in haec verba herein. It is also noted that the terms “comprising” and “containing” are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the present disclosure, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

(146) This application refers to various journal articles, and other publications, all of which are incorporated herein by reference. All references, patents and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document. If there is a conflict between any of the incorporated references and the instant specification, the specification shall control. In addition, any particular embodiment of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the present disclosure can be excluded from any claim, for any reason, whether or not related to the existence of prior art.

EXAMPLES

Example 1

Cloning of *M. thermophile* CYP505A30 and *E. coli* Icd

(147) The amino acid sequence of a BM3 homolog of *Myceliophthora thermophile* (CYP505A30) was obtained from UniProt (www.uniprot.org/uniprot/G2QDZ3.fasta) and the corresponding gene was codon optimized for expression in *Escherichia coli* and synthesized by GenScript (Piscataway, NJ). The nucleotide sequence of CYP505A30 from *M. thermophile* with codon-optimization for *E. coli* expression is provided as SEQ ID NO: 1. The amino acid sequence of CYP505A30 is provided as SEQ ID NO: 3.

(148) The resulting gene product was cloned into pETDuet-1 vector (AMP.sup.+, Novagen)

through NdeI and XhoI sites. The construct was transformed into BL21 (DE3) cells for expression. To enhance CYP505A30 activity in *E. coli*, an isocitrate dehydrogenase (Icd)-mediated NADPH regenerating system was coupled with CYP505A30. Primers ICD-F, 5'-ggaattcCATatgGAAAGTAAAGTAGTTGT (SEQ ID NO: 7) and ICD-R, 5'-ccgCTCGAGttaCATGTTTTTCGATGATCGCGT (SEQ ID NO: 8) were used to amplify the *icd* gene by PCR from the genome of *E. coli* K-12 substr. MG1655. The nucleotide sequence of *icd* from *E. coli* K-12 substr. MG1655 is provided as SEQ ID NO: 2. The corresponding amino acid sequence is provided as SEQ ID NO: 4.

(149) The resulting PCR product was subcloned into pCDFDuet-1 vector (Spect.sup.+, Novagen) through NdeI and XhoI sites. The construct was used to transform BL21 (DE3) cells overexpressing CYP505A30 and the transformants were selected on LB (AMP.sup.+Spect.sup.+) plates.

(150) The resulting vectors are used to overexpress *M. thermophila* CYP505A30 and *E. coli* Icd in *E. coli* cells.

Example 2

Production of Gamma- and Delta-Lactones Using *E. coli* Cells Overexpressing *M. Thermophila* CYP505A30 and *E. coli* Icd

(151) This example demonstrates the use of *E. coli* cells overexpressing *M. thermophila* CYP505A30 and *E. coli* Icd in the bioconversion of short chain fatty acids into gamma- and delta-lactones via a subterminal hydroxy fatty acid intermediate.

(152) To produce subterminal (i.e., omega-1 (ω -1), omega-2 (ω -2), and omega-3 (ω -3)) hydroxy fatty acids from corresponding short chain fatty acids, *E. coli* cells overexpressing CYP505A30 and Icd were incubated with medium comprising the short chain fatty acids as substrates.

(153) In a typical experiment, an overnight culture was used to inoculate liquid LB medium (2%) containing 100 mg/L of carbenicillin and 100 mg/L of spectinomycin. The culture was first grown at 37° C. to an OD_{sub}600 of 0.6 and cooled down to 16° C. Then 1 mM IPTG was added to induce protein expression. After 16 h of incubation at 16° C., cells were harvested by centrifugation.

(154) Harvested cell pellets were re-suspended at a concentration of 100 g/L fresh weight in 100 mM potassium phosphate buffer (pH7.0) containing 0.1% Tween 40. Then 3 g/L of lauric acid (C12:0) or 3 g/L of tridecanoic acid (C13:0) together with 10 g/L of isocitric acid trisodium salt were added for biotransformation at 30° C. in a shaker. Samples were taken at different reaction times and analyzed by GC/FID.

(155) GC/FID analysis was conducted on Shimadzu GC-2014 system. The analytical column is Restek RXi-5 ms (thickness 0.25 μ m; length 30 m; diameter 0.25 mm) and the injection temperature is 240° C. under split mode. The temperature gradient is 0-3 min, 100° C.; 3-9 min 100° C.-280° C., a gradient of 30; 9-12 min, 280° C.

(156) As shown in FIG. 2A and FIG. 2B, HO-C12:0 and HO-C13:0 were efficiently produced from the corresponding carboxylic acids C12:0 (dodecanoic acid, also known as lauric acid) and C13:0 (tridecanoic acid, also known as tridecylic acid) by *E. coli* cells modified according to the procedures described in Example 1. According to Baker et al. (2017), CYP505A30 catalyzed hydroxylation of straight chain fatty acids at the ω -1 to ω -3 positions. However, FIGS. 2A-2B show single peaks because GC/FID cannot separate ω -1 to ω -3 hydroxy fatty acids in the absence of derivatization. As a result, the peaks represent a mixture of ω -1 to ω -3 hydroxy fatty acids.

(157) These results evidence production of ω -1 to ω -3 hydroxy fatty acids from corresponding fatty acids by re-suspended *E. coli* cells overexpressing *M. thermophila* CYP505A30 and *E. coli* Icd.

(158) To produce gamma-lactones (5-membered ring) and delta-lactones (6-membered ring) from subterminal hydroxy fatty acids, the mixture of ω -1 to ω -3 hydroxy fatty acids from the previous step was first isolated from the re-suspended *E. coli* cells then inoculated with a *Candida boidinii* culture.

(159) As an example of producing lactones from hydroxy fatty acids, the mixture of HO-C12:0 and the mixture of HO-C13:0, respectively, was mixed with 60 g/L of corn steep liquor (CSL) at a 1:1 ratio, and the resulting medium was autoclaved at 121° C. for 20 min. The media was named as CYP505A30C12 or CYP505A30C13 for C12:0 and C13:0 bioconversion, respectively. Then the medium was inoculated with an overnight yeast culture (e.g., *Candida boidinii*), and the yeast culture was grown in a 30° C. shaker. Samples were taken daily after inoculation and analyzed as described herein.

(160) To analyze γ -lactone production, 0.5 ml *E. coli* culture was taken and 10 μ l 2N HCl was added. The acidified culture was extracted with 0.5 ml ethyl acetate with shaking at room temperature for 60 min. After centrifugation at 14,000 rpm for 15 min, the ethyl acetate phase was used for GC/MS or GC/FID analysis.

(161) GC/MS analysis was conducted on Shimadzu GC-2010 system coupled with GCMS-QP2010S detector. The analytical column is SHRXI-5 MS (thickness 0.25 μ m; length 30 m; diameter 0.25 mm) and the injection temperature is 265° C. under split mode. The temperature gradient is 0-3 min 80° C.; 3-8.7 min 120° C. to 263° C., a gradient of 25; 8.7-10.7 min, 263° C.

(162) As shown in FIG. 3, if the initial starting fatty acid has an even number of carbons such as C12:0 (dodecanoic acid, also known as lauric acid), after β -oxidation in yeast, the predicted products would be δ -hexalactone (DC6) from the ω -1 hydroxy fatty acid intermediate, γ -hexalactone (GC6) from the ω -2 hydroxy fatty acid intermediate, and δ -octalactone (DC8) from the ω -3 hydroxy fatty acid intermediate. This is because in each round of β -oxidation, 2 carbon atoms are removed from the hydroxy fatty acid intermediate. Depending on the position of the OH group in the hydroxy fatty acid, there can be 9, 8, or 7 intervening carbon atoms between the hydroxylated carbon atom and the carboxylated carbon atom in the hydroxy fatty acid. Once β -oxidation has removed enough pairs of carbon atoms such that there are either 2 or 3 intervening carbon atoms remaining, favorable kinetics lead to cyclization into the lactone product (a γ -lactone if there are 2 intervening carbon atoms remaining, and a δ -lactone if there are 3 intervening carbon atoms remaining).

(163) FIG. 4 shows the spectra from GC/MS analysis obtained with CYP505A30C12 (containing ω -1 to ω -3 hydroxy lauric acid) following incubation with *Candida boidinii*. Referring to FIG. 4A, peaks 1, 2, 3, and 4 were analyzed. After comparing the retention times to those of known standards, peak 1 was identified to be the γ -hexalactone GC6, peak 2 was identified to be the δ -hexalactone DC6, and peak 4 was identified to be the δ -octalactone DC8. Peak 3 was identified to be phenethyl alcohol, which is a metabolite of yeast metabolism.

(164) As shown in FIG. 5, if the initial starting fatty acid has an odd number of carbons such as C13:0 (tridecanoic acid, also known as tridecylic acid), after β -oxidation in yeast, the predicted products would be γ -valerolactone (GC5) from the ω -1 hydroxy fatty acid intermediate, δ -heptalactone (DC7) from the ω -2 hydroxy fatty acid intermediate, and γ -heptalactone (GC7) from the ω -3 hydroxy fatty acid intermediate. This is because in each round of β -oxidation, 2 carbon atoms are removed from the hydroxy fatty acid intermediate. Depending on the position of the OH group in the hydroxy fatty acid, there can be 10, 9, or 8 intervening carbon atoms between the hydroxylated carbon atom and the carboxylated carbon atom in the hydroxy fatty acid. Once β -oxidation has removed enough pairs of carbon atoms such that there are either 2 or 3 intervening carbon atoms remaining, favorable kinetics lead to cyclization into the lactone product (a γ -lactone if there are 2 intervening carbon atoms remaining, and a δ -lactone if there are 3 intervening carbon atoms remaining).

(165) FIG. 6 shows the spectra from GC/MS analysis obtained with CYP505A30C13 (containing ω -1 to ω -3 hydroxy tridecanoic acid) following incubation with *Candida boidinii*. Referring to FIG. 6A, peaks 1, 2, 3, and 4 were analyzed. After comparing the retention times to those of known standards, peak 2 was identified to be the γ -heptalactone GC7, and peak 3 was identified to be the δ -heptalactone DC7. Peak 1 was identified to be phenethyl alcohol, which is a metabolite of yeast

metabolism. Valerolactone (GC5) was not detected by GC/MS most likely due to its low concentration or volatility.

(166) These results demonstrate production of lactones from hydroxy fatty acids by inoculation with a yeast cell culture.

Example 3

Production of Lactones Using *E. coli* Cells Overexpressing *U. Isabellina* Cytochrome P450 Protein

(167) The amino acid sequence of a BM3 homolog from *Umbelopsis isabellina*, a cytochrome P450 protein, was obtained from JGI Genome Portal (www.genome.jgi.doe.gov/cgi-bin/dispgeneModel?db=Umbisal&id=523695) and the corresponding gene was codon-optimized for expression in *Escherichia coli* and synthesized by Gene Universal Inc. (Newark, DE). The nucleotide sequence of BM3 homolog of *Umbelopsis isabellina*, MI2, with codon-optimization for *E. coli* expression is provided below as SEQ ID NO: 5. The amino acid sequence of MI2 is provided below as SEQ ID NO: 6.

(168) The resulting gene product was cloned into pET-32a(+) vector (AMP⁺, Novagen) through HindIII and XhoI sites. The construct was transformed into BL21 (DE3) cells for expression, and an overnight culture was used to inoculate liquid LB medium (2%) containing 100 mg/L of carbenicillin. After 16 h of incubation at 16° C., cells were harvested by centrifugation.

(169) Harvested cell pellets were resuspended at a concentration of 100 g/L fresh weight in 100 mM potassium phosphate buffer (pH7.0) containing 0.1% Tween 40. Then 1 g/L of lauric acid (C12:0) or 1 g/L of tridecanoic acid (C13:0) was added for biotransformation at 30° C. in a shaker. Samples were taken at different reaction times and the production of hydroxy fatty acids were determined by GC/MS after silylation with MSTFA (N-methyl-N-trimethylsilyl-trifluoroacetamide).

(170) FIG. 7A and FIG. 7B demonstrate production of HO-C12:0 and HO-C13:0 from the corresponding carboxylic acids by *E. coli* cells overexpressing BM3 homolog of *Umbelopsis isabellina*, MI2. Although a mass spectrum library search did not find exact matches for peak 1 in the samples, the search indicated that the peak was hydroxy fatty acids.

(171) To produce lactones from the hydroxy fatty acids HO-C12:0 and HO-C13:0, the biotransformation mixture was mixed with 60 g/L of corn steep liquor (CSL) at 1:1 ratio, and the resulting medium was autoclaved at 121° C. for 20 min. The media was labelled as MI2-C12-CSL or MI2-C13-CSL for C12:0 and C13:0 bioconversion, respectively. Then the medium was inoculated with an overnight yeast culture, e.g., *Candida boidinii* and the yeast culture was grown in a 30° C. shaker. Samples were taken daily after inoculation.

(172) To analyze lactone production in yeast cultures, 0.5 ml *Candida boidinii* culture was taken and 10 µl 2N HCl was added. The acidified culture was extracted with 0.5 ml ethyl acetate with shaking at room temperature for 60 min. After centrifugation at 14,000 rpm for 15 min, the ethyl acetate phase was used for GC/MS or GC/FID analysis.

(173) FIG. 8A and FIG. 8B demonstrate that GC6 is the major lactone derived from C12:0 (via HO-C12:0 as an intermediate) and DC7 is the major lactone derived from C13:0 (via HO-C13:0 as an intermediate), respectively. This suggests that ω-2 hydroxy lauric acid and ω-2 hydroxy tridecanoic acid likely correspond to peak 1 in FIG. 7A and FIG. 7B, respectively, which in turn were derived from C12 and C13, respectively.

(174) Taken together, these results demonstrate that *Umbelopsis isabellina* cytochrome P450 MI2 is an ω-2 fatty acid hydroxylase and can be used for production of γ-hexalactone or 8-heptalactone.

(175) TABLE-US-00001 Sequences of Importance *M. thermophile* CYP505A30 nucleotide sequence (SEQ ID NO: 1)

```
ATGGCGGATAAGACCACCGAAACCGTGCCGATTCCGGGTCCGCCGGGCCTG
CCGCTGGTTGGTAATGCGCTGGCGTTTGATAGCGAACTGCCGCTGCGTACC
TTCCAGGAATTTGCGGAGGAATACGGCGAGATCTATCGTCTGACCCTGCCG
ACCGGTACCACCCTGGTGGTTAGCAGCCAAGCGCTGGTTCACGAACTGTGC
```

ACGATAAGCGTGTTCAAGAAAGCGCGGTTGCTGCGGCGCTGGCGGAAGTTGCGT
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GGCATCGCGCACCGTATTCTGATGCCGGCGTTTGGTCCGGCGAGCATTACAG GGC
ATGTTTACCGAAATGCACGAGATCGCGAGCCAACTGGCGCTGAAATGGGCG
CGTCACGGTCCGGACACCCCGATTTTCGTTACCGACGATTTTACCCGTCTG
ACCCTGGATAACCCTGGCGCTGTGCACCATGAACTTCCGTTTTAACAGCTAC
TATCACGACGAACTGCACCCGTTTCATCAACGCGATGGGCAACTTTCTGACC
GAGAGCGGTGCGCGTGCGATGCGTCCGGCGATCACCAGCATTTTCCACCAG
GCGGCGAACCCTAAGTACTGGGAAGATATTGAGGTTCTGCGTAAAACCGCG
CAAGGTGTGCTGGACACCCGTCGTAAGCACCCGACCAACCGTAAAGATCTG
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GCGTGATCGGTTCGTCCGTTTTGCGTGGCAGGAAGCGGTGCTGGTTATGGCG
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CACTACAAGCAGACCCTGACCACCAAGCCGAAAGACTTCTATATGCGTGCG
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GCGACCGAGAAGCTGCCGACCGATCGTCCGGTGGTTATCATTACCGCGAGC
TTCGAGGGTCAGCCGCCGGACAACGCGGCGAAGTTTTGCGGCTGGCTGAAA
AACCTGGAAGGTGATGAGCTGAAAAACGTGAGCTACGCGGTTTTTCGGTTGC
GGCCACCACGACTGGAGCCAGACCTTTCACCGTATTCCGAAGCTGGTTTAC
CAAACCATGAAAGCGCACGGTGCGAGCCCGATCTGCGACGAAGGCCTGACC
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GTGTTCTGGCCGGCGGTTTCGTGCGCGTTATGGCGCGGCGGGTGCGGTTGCG
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GACAAGCTGCTGGTGAGCGTTTCGTCCGACCCACACCGCGTTTTCTGCTGCCG

GACCAAGATAAACCACCGCATATTGCGTGCGGTAGCGGTCTGGCGCCGTTCCGTGGTTTTATCCAGGAACGTGCGGGCGCTGCTGGCGAAAGGTACCCAACCTGGCGGGCGGCGCTGCTGTTCTACGGTTGCCGTAGCCCCG
GAGAAGGACGATCTGTATCGTGACGAATTCGATAAATGGCAAGAGAGCGGTGCGGTGGATGTTTCGTTCGTGCGTTTAGCCGTGTTGATAGCGACGATAACCGAGGCGCGTGTTGCCGTACGTTTCAGGACCGTCTGTGGCACGATCGTGAAGAGGTGAAGGCGCTGTGGGACCGTGGCGCGCGTGTGTACGTTTGCGGTAGCCGT
CAAGTGGGCGAAGGTGTTAAAACCGCGATGGGCCGTATCGTGCTGGGTGAA
GAGGACGCGGAGGATGCGATCAGCAAGTGGTATGAAACCGTGCGTAATGACCGTTATGCGACCGATGTGTTTCGACTAA

icd gene from *E. coli* K-12 substr. MG1655 (SEQ ID NO: 2)

GAAAGTAAAGTAGTTGTTCCGGCACAAGGCAAGAAGATCACCTGCAAAACGGCAAACCTCAACGTTCCCTGAAAATCCGATTATCCCTTACATTGAAGGTGATGGAATCGGTGTAGATGTAACCCCAGCCATGCTGAAAGTGGTCGACGCTGCA
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GAGATCGTCATTAAAGACGTGATTGCTGATGCATTCCTGCAACAGATCCTGCTGCGTCCGGCTGAATATGATGTTATCGCCTGTATGAACCTGAACGGTGAC
TACATTTCTGACGCCCTGGCAGCGCAGGTTGGCGGTATCGGTATCGCCCCTGGTGCAAACATCGGTGACGAATGCGCCCTGTTTGAAGCCACCCACGGTACT
GCGCCGAAATATGCCGGTCAGGACAAAGTAAATCCTGGCTCTATTATTCTCTCCGCTGAGATGATGCTGCGCCACATGGGTTGGACCGAAGCGGCTGACTTA
ATTGTTAAAGGTATGGAAGGCGCAATCAACGCGAAAACCGTAACCTATGAC
TTCGAGCGTCTGATGGATGGCGCTAAACTGCTGAAATGTTTCAGAGTTTGGTGACGCGATCATCGAAAACATGTAA

M. thermophile CYP505A30 amino acid sequence (SEQ ID NO: 3)

MADKTTETVPIPGPPGLPLVGNALAFDSELPLRTFQEFAEEYGEIYRLTLP
TGTTLVVSSQALVHELCDKRFKKPVAAALAEVRNGVNDGLFTAREEEPNW
GIAHRILMPAFGPASIQGMFTEMHEIASQLALKWARHGPDTPIFVTDDFTR
LTLDTLALCTMNFREFNSYYHDELHPFINAMGNFLTESGARAMRPAITSIFH
QAANRKYWEDIEVLRKTAQGVLDTRRKHPTNRKDLLSAMLGDGVDAGTQKL
SDSSIIDNLITFLIAGHETTSGLLSFAFYLLIKHQDAYRKAQEEVDRVIGK
GPIKVEHIKKLPYIAAVLRETLRLCPTIPIINRAAKQDEVIGGKYAVAKDQ
RLALLLAQSHLDPVYGETAKQFIPERMLDENFERLNREYPDCWKPFGTGM
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GDAIENM *Umbelopsis isabellina* MI2 nucleotide sequence (SEQ ID NO: 5)
ATGAGCACCACCACCCTGATTGTGGCCATTAAAGGGTACCGATAATGTTGAA
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ATTCTGGAAGATGCAAATGGTGACAATCTGACCACCATTGATCAGGTTCTG
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CTGGGCTTTGGTCTGATTCTGGATAGTGTGTTTGAAATTACCAGCGTGAT
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CCGATTCGTGCACGTATTAAGAGTAGCAAAAGTACCTTTGGCCTGCCGGAA
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CATCTGTTTTTTCGGCTGCCGCCATCCGGATGAAGATTTTATCTATCGTAGC
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TATGTTTGCGGCAGTGGCCCGATGAGCCGCGATGTGCGTCGTAGCTTTGAA
CTGATGGCAAAAAGTTTTTGAGGGCGCCAAAACCGATGATGAAGCATGCGAA
AAACTGCTGGAATTCATGAGCGCAGGTCGTTATAATGAAGATGTGTGGGGT TAA

Umbelopsis isabellina MI2 amino acid sequence (SEQ ID NO: 6)
MSTTTLIVAIKGTDNVERDGAIVRLDAGASMETLRPRIAEKLAISSGIEDL
ILEDANGDNLTTIDQVRKQQT V FVNLEDQIKLPVPAHTLPYFGNLYQLLP
DMLAGWRKLFDEYGPVVKVNLLGNEIIGTNDPAVAELWVKESEYFTKKIYG
GLQEVKSFGGQGLFTTDSDDMDWKL AHKLLMPAFSPRAIKVYQHEMSVIGL
QTIKVFEQYSPDEEVEILHWTTNLT FETIGKVGFYDFHLLDDRYGENHPF
IEAMGYCMKQSFARGTQSKLIKYLPIEANRRYDRSLNLMHSIVDEVITQRK
SHPHASEDNKDLLDFMLTARDENNLGLSDENIRDQVITFLIAGHETTSNTL
AWTLYELSRHPEIEQKILQE VVNLGITTD SLPTSEQSSSMKYTYQVLKETL
RMYSPLRALAKYCKKDIVVPGGYQIKAGDRVAVQLNSLHYNEKVYPNPQTQY
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QMAMPTNATPVTAFLPQVGTFTYASQLASQAKSFGFKDITL
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 TSKDNDLPLKGLLYTVFGLGNRNWRTYQQFPIKCDARLDELGGERFYDLGS
 GNADKMDGFEHWDCAHFWTHTLTYYGIATHEGSASLVPGQKTTEPTSGL
 EIRFIPPSESEAWDKADKNVNGDYNAEIQVNRELQNIERSKRSTRHIEIDI
 SKLQSPDTDKHYRLTGDHLEVYPENADNVVESIALGFGILDSVFEITSVD
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 VAASFSDVIIPGENGQKAYNEFIQKYRNLLDLQRGFPLKELDLKEFLCAVS
 VMQPRRYSIASSPLKAEDTAYLAVGVVDDVFANRHYYGLASGYLARSQPPT
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 HLFFGCRHPDEDFIYRSVFEGYEKSGVITKLYPAFSRLGEDNPRKYVQHQL
 MANAGKIWTLLTNGANVYVCGSGPMSRDVRRSFELMAKSFGGAKTDDEACE
 KLEFMSAGRYNEDVWG ICD-F Primer (SEQ ID No: 7)
 GGAATTCCATATGGAAAGTAAAGTAGTTGT ICD-R Primer (SEQ ID No: 8)
 CCGCTCGAGTTACATGTTTTTCGATGATCGCGT

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TABLE 1	Examples of fatty acids	Common Name	Systematic Name
Structural Formula	Lipid Numbers	Valeric acid	Pentanoic acid
CH.sub.3(CH.sub.2).sub.3COOH	C5:0	Caproic acid	Hexanoic acid
CH.sub.3(CH.sub.2).sub.4COOH	C6:0	Enanthic acid	Heptanoic acid
CH.sub.3(CH.sub.2).sub.5COOH	C7:0	Caprylic acid	Octanoic acid
CH.sub.3(CH.sub.2).sub.6COOH	C8:0	Pelargonic acid	Nonanoic acid
CH.sub.3(CH.sub.2).sub.7COOH	C9:0	Capric acid	Decanoic acid
CH.sub.3(CH.sub.2).sub.8COOH	C10:0	Undecylic acid	Undecanoic acid
CH.sub.3(CH.sub.2).sub.9COOH	C11:0	Lauric acid	Dodecanoic acid
CH.sub.3(CH.sub.2).sub.10COOH	C12:0	Tridecylic acid	Tridecanoic acid
CH.sub.3(CH.sub.2).sub.11COOH	C13:0	Myristic acid	Tetradecanoic acid
CH.sub.3(CH.sub.2).sub.12COOH	C14:0	Pentadecylic acid	Pentadecanoic acid
CH.sub.3(CH.sub.2).sub.13COOH	C15:0	Palmitic acid	Hexadecanoic acid
CH.sub.3(CH.sub.2).sub.14COOH	C16:0	Margaric acid	Heptadecanoic acid
CH.sub.3(CH.sub.2).sub.15COOH	C17:0	Stearic acid	Octadecanoic acid
CH.sub.3(CH.sub.2).sub.16COOH	C18:0	Nonadecylic acid	Nonadecanoic acid
CH.sub.3(CH.sub.2).sub.17COOH	C19:0	Arachidic acid	Eicosanoic acid
CH.sub.3(CH.sub.2).sub.18COOH	C20:0	Heneicosylic acid	Heneicosanoic acid
CH.sub.3(CH.sub.2).sub.19COOH	C21:0	Behenic acid	Docosanoic acid
CH.sub.3(CH.sub.2).sub.20COOH	C22:0	Tricosylic acid	Tricosanoic acid
CH.sub.3(CH.sub.2).sub.20COOH	C23:0	Lignoceric acid	Tetracosanoic acid
CH.sub.3(CH.sub.2).sub.22COOH	C24:0	Pentacosylic acid	Pentacosanoic acid
CH.sub.3(CH.sub.2).sub.23COOH	C25:0	Cerotic acid	Hexacosanoic acid
CH.sub.3(CH.sub.2).sub.24COOH	C26:0	Heptacosylic acid	Heptacosanoic acid
CH.sub.3(CH.sub.2).sub.25COOH	C27:0	Montanic acid	Octacosanoic acid
CH.sub.3(CH.sub.2).sub.26COOH	C28:0	Nonacosylic acid	Nonacosanoic acid
CH.sub.3(CH.sub.2).sub.27COOH	C29:0	Melissic acid	Triacontanoic acid
CH.sub.3(CH.sub.2).sub.28COOH	C30:0	Henatriacontylic acid	Henatriacontanoic acid
CH.sub.3(CH.sub.2).sub.29COOH	C31:0	Lacceroic acid	Dotriacontanoic acid
CH.sub.3(CH.sub.2).sub.30COOH	C32:0	Psyllic acid	Trtriacontanoic acid
CH.sub.3(CH.sub.2).sub.31COOH	C33:0	Geddic acid	Tetratriacontanoic acid
CH.sub.3(CH.sub.2).sub.32COOH	C34:0	Ceroplactic acid	Pentatriacontanoic acid
CH.sub.3(CH.sub.2).sub.33COOH	C35:0	Hexatriacontylic acid	Hexatriacontanoic acid
CH.sub.3(CH.sub.2).sub.34COOH	C36:0	Heptatriacontanoic acid	Heptatriacontanoic acid
CH.sub.3(CH.sub.2).sub.35COOH	C37:0	Octatriacontanoic acid	Octatriacontanoic acid
CH.sub.3(CH.sub.2).sub.36COOH	C38:0	Nonatriacontanoic acid	Nonatriacontanoic acid

CH.sub.3(CH.sub.2).sub.37COOH C39:0 Tetracontanoic acid Tetracontanoic acid
 CH.sub.3(CH.sub.2).sub.38COOH C40:0
 (177) TABLE-US-00003 TABLE 2 Sequence Information Seq ID No. Sequence
 Description 1 *M. thermophile* CYP505A30 nucleotide sequence with codon-
 optimization for *E. coli* 2 *icd* gene from *E. coli* K-12 substr. MG1655 3 *M.*
thermophile CYP505A30 amino acid sequence 4 *icd* protein from *E. coli* K-12
 substr. MG1655 5 *Umbelopsis isabellina* MI2 BM3 homolog nucleotide sequence codon-
 optimization for *E. coli* expression 6 *Umbelopsis isabellina* MI2 BM3 homolog
 amino acid sequence 7 ICD-F Primer 5'GGAATTCCATATGGAAAGTAAAGTAGTTGT3' 8
 ICD-R Primer 5'CCGCTCGAGTTACATGTTTTTCGATGATCGCGT3'

Claims

1. A method of making lactones, the method comprising: (a) incubating a cellular system expressing a heterologous cytochrome P450 (CYP450) protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, wherein the straight chain fatty acid and the subterminal hydroxy fatty acid have 5 to 40 carbon atoms, and wherein the heterologous cytochrome P450 protein comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of a cytochrome P450 protein from *Umbelopsis isabellina*, and either: (b1) incubating the subterminal hydroxy fatty acid with a yeast cell culture to produce lactones, wherein the lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof, or (b2) incubating the subterminal hydroxy fatty acids in an acidic environment to produce lactones, wherein the lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof.
2. The method of claim 1, wherein the heterologous cytochrome P450 protein comprises the amino acid sequence of SEQ ID NO: 6.
3. The method of claim 1, wherein the cellular system expresses an exogenous isocitrate dehydrogenase (*Icd*) protein besides expressing an heterogenous cytochrome 450 protein.
4. The method of claim 3, wherein the *Icd* protein comprises an amino acid sequence that is at least 95% identical to the amino acid sequence of an *Icd* protein from *E. coli*.
5. The method of claim 3, wherein the *Icd* protein comprises the amino acid sequence of SEQ ID NO: 4.
6. The method of claim 1, wherein the straight chain fatty acid and the hydroxy fatty acid have 7 carbons to 20 carbons.
7. The method of claim 1, wherein the straight chain fatty acid is lauric acid or tridecylic acid.
8. The method of claim 1, wherein the subterminal hydroxy fatty acid is a hydroxylated lauric acid, or a hydroxylated tridecylic acid.
9. The method of claim 1, wherein the cellular system comprises bacterial cells, yeast cells, or a combination thereof.
10. The method of claim 1, wherein the lactones comprise five carbons to eight carbons.
11. The method of claim 1, wherein the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone.
12. The method of claim 1, wherein the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.
13. The method of claim 1, wherein the straight chain fatty acid is lauric acid, the subterminal hydroxyl fatty acid comprises one or more of ω -1 hydroxy lauric acid, ω -2 hydroxy lauric acid, and ω -3 hydroxy lauric acid, and the lactones comprise one or more of δ -hexalactone, γ -hexalactone, and δ -octalactone.
14. The method of claim 1, wherein the straight chain fatty acid is tridecylic acid, the subterminal hydroxy fatty acid comprises one or more of ω -1 hydroxy tridecylic acid, ω -2 hydroxy tridecylic

acid, and ω -3 hydroxy tridecylic acid, and the lactones comprise one or more of γ -valerolactone, δ -heptalactone, and γ -heptalactone.

15. The method of claim 1, wherein the subterminal hydroxy fatty acid is isolated from the cellular system prior to step (b1) or (b2).

16. The method of claim 1, further comprising isolating the lactones from the yeast cell culture.

17. A method of making lactones, the method comprising: (a) incubating a cellular system expressing a heterologous cytochrome P450 protein with a medium comprising a straight chain fatty acid to produce a subterminal hydroxy fatty acid, wherein the heterologous cytochrome P450 protein comprises an amino acid sequence that is at least 90% identical to the amino acid sequence of a cytochrome P450 protein from *Umbelopsis isabellina*; and wherein the straight chain fatty acid and the subterminal hydroxy fatty acid have 5 to 40 carbon atoms; (b) incubating the subterminal hydroxy fatty acid from step (a) with a cell culture to produce subterminal hydroxy fatty acid with the length shorter than the subterminal hydroxy fatty acid from step (a); and (c) incubating the subterminal hydroxy fatty acids obtained at the end of step (b) in an acidic environment to produce lactones, wherein the lactones comprise at least one gamma-lactone, at least one delta-lactone, or combinations thereof.
